THE HYDROGEN ATOM

General Chemistry I, Lecture Series 6

Qixi Mi

Reading:
OGB8 §5.1

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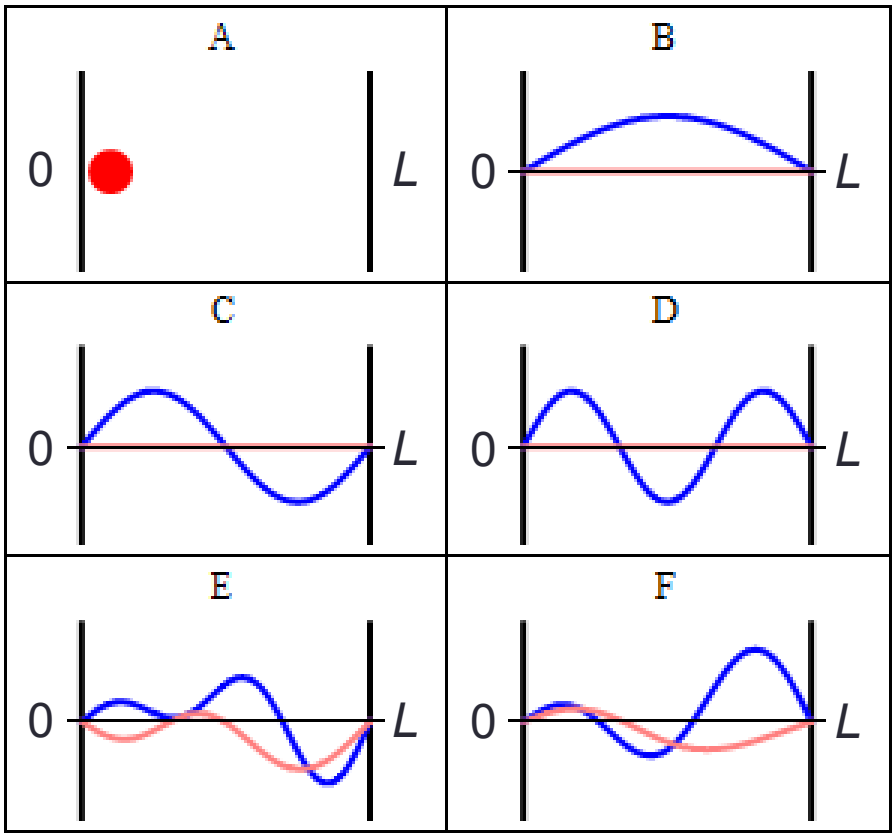
Outline

- Oscillations in 2D
- Atomic orbitals: Appearance
- Atomic orbitals: Properties

1D Standing Wave (1)

(1) $\psi(x) = 0$ for $x \leq 0$ or $x \geq L$ (Boundary condition 边界条件)

(2) $\int |\psi(x)|^2 \, dx = 1$

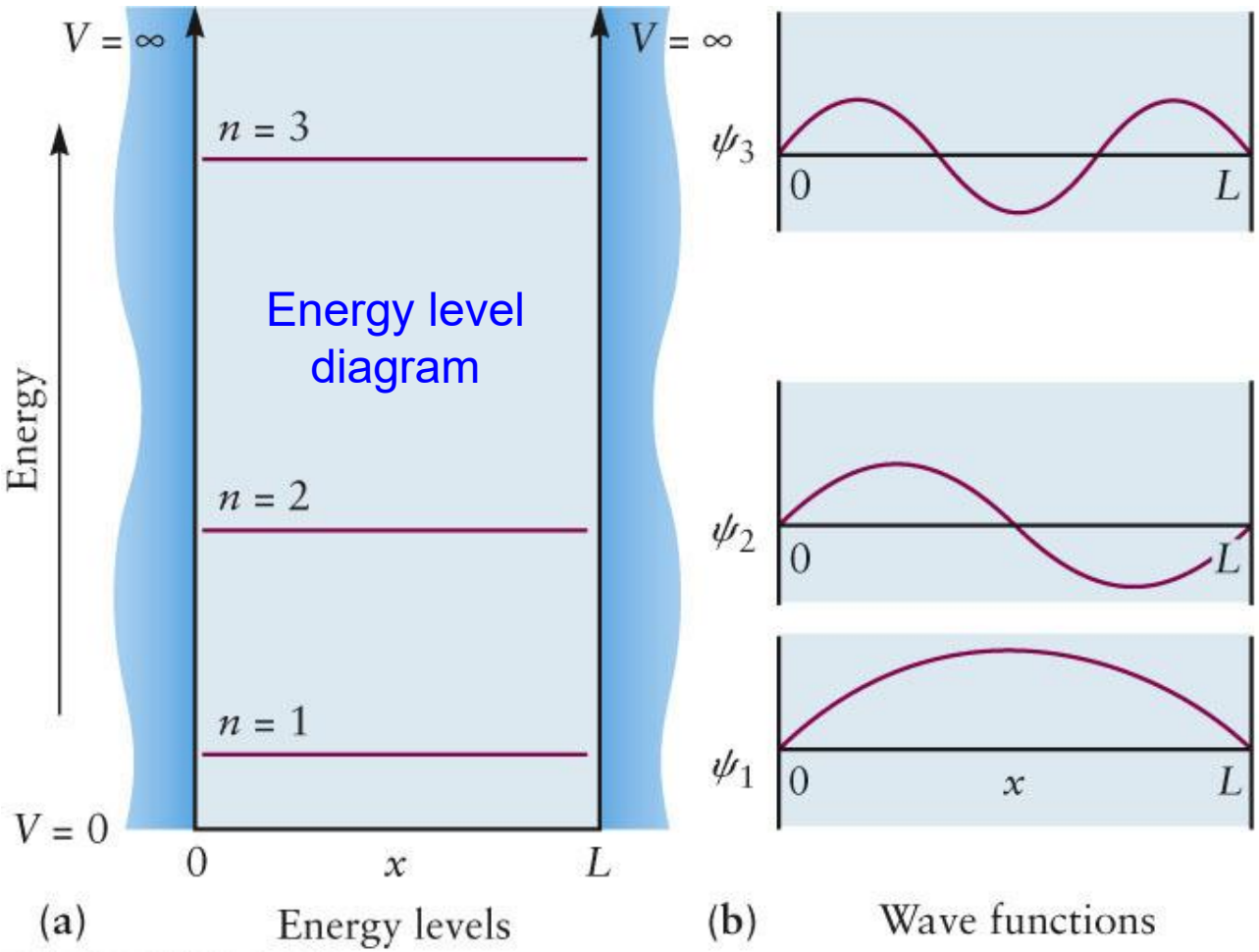


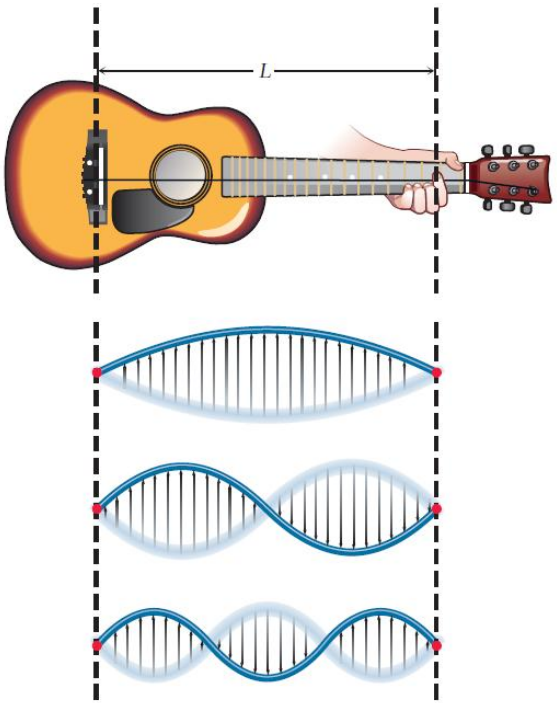
Stationary states 定态

Time-dependent state

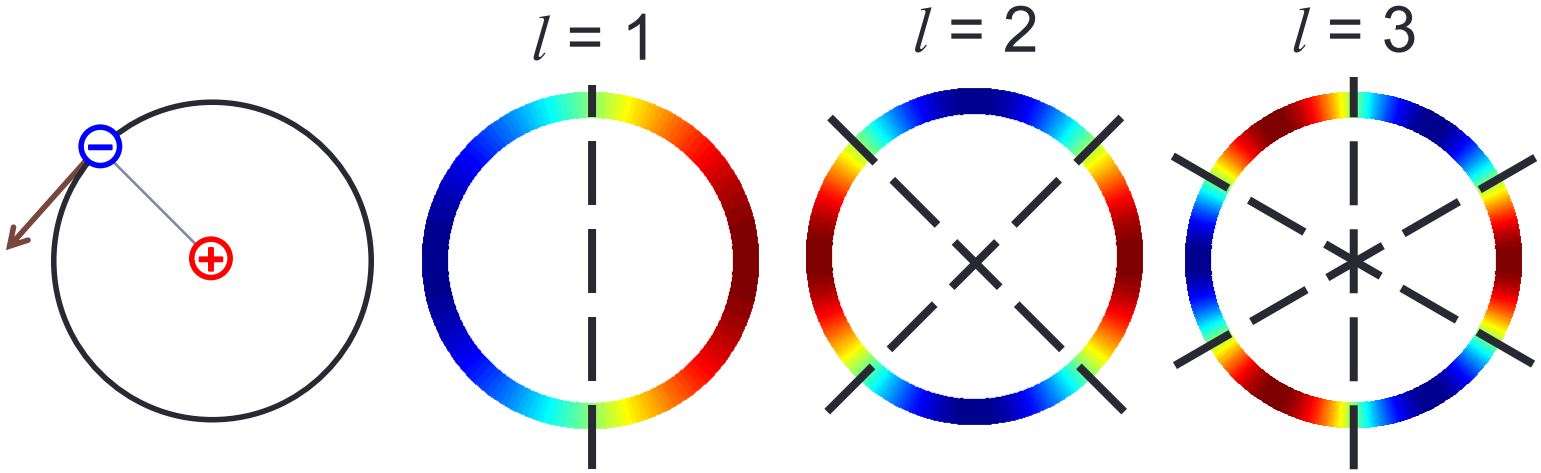
1D Standing Wave (2)

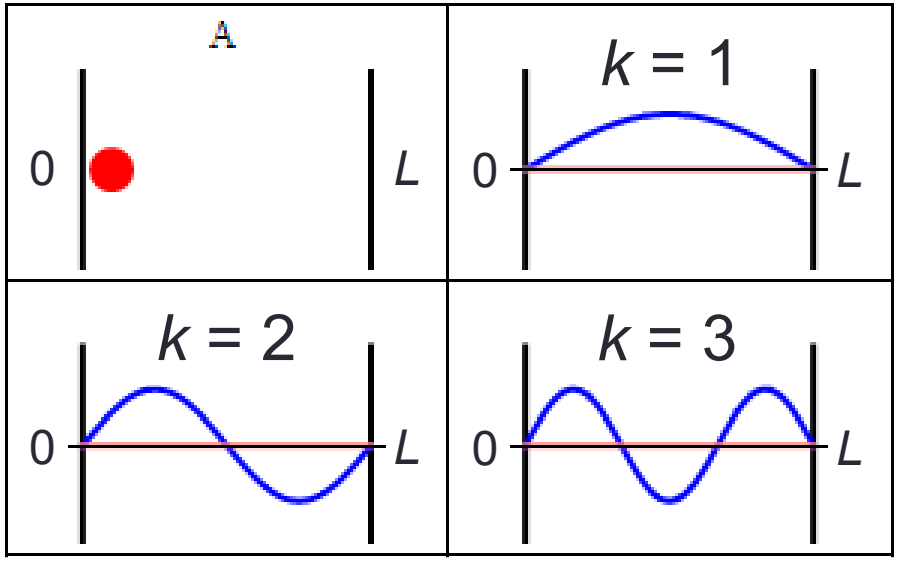
The nth Wave function: $\psi_n(x) \propto \sin\left(n\pi\frac{x}{L}\right)$, n = number of nodes + 1.

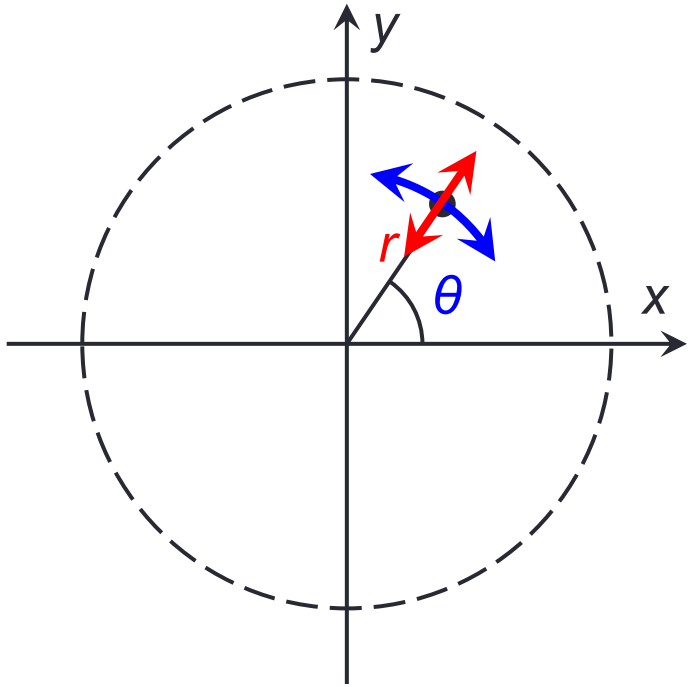




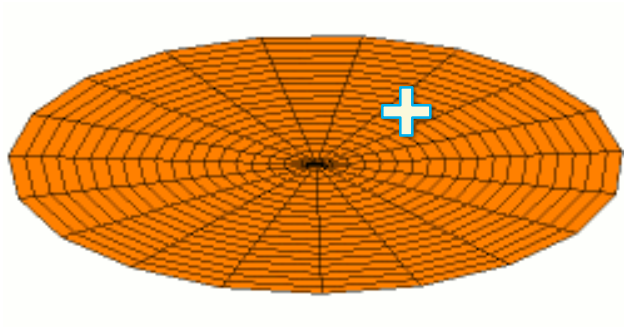
From 1D to 2D Standing Waves







The 1*s* Wave



主量子数
Principal quantum number n = Total number of nodes + 1

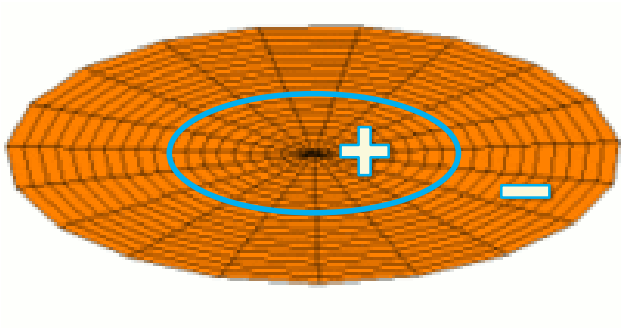
角(动量)量子数
Angular quantum number l = Number of angular nodes

$l = 0, 1, 2, 3\ldots \Longrightarrow s, p, d, f\ldots$

No nodes.

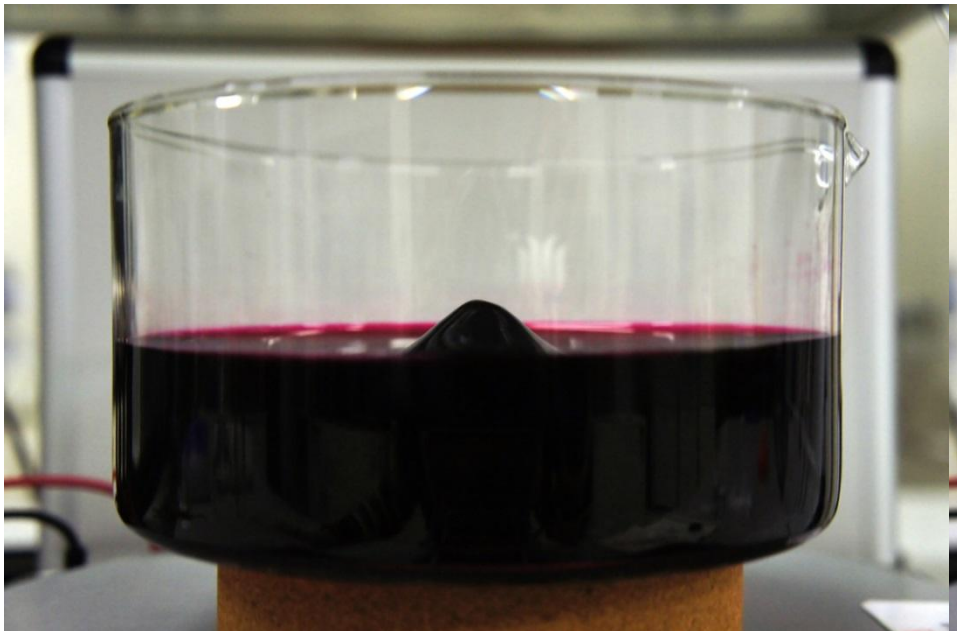
$n = 1, l = 0 \Longrightarrow 1s$

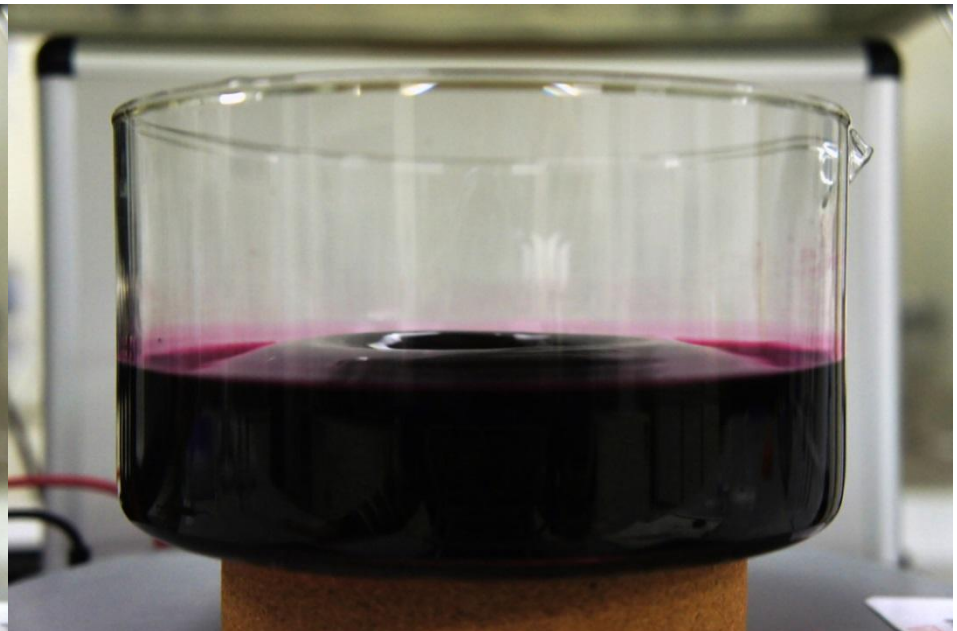
The 2*s* Wave



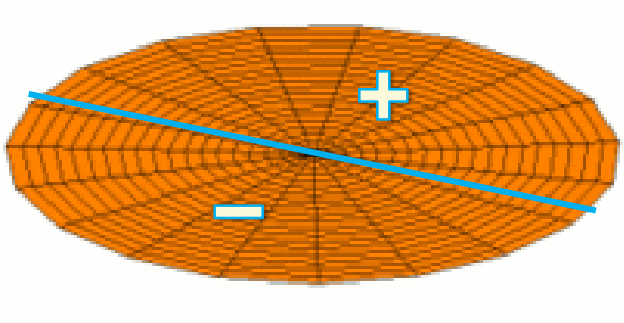
1 nodal ring (radial node)

$n = 2, l = 0 \Longrightarrow 2s$



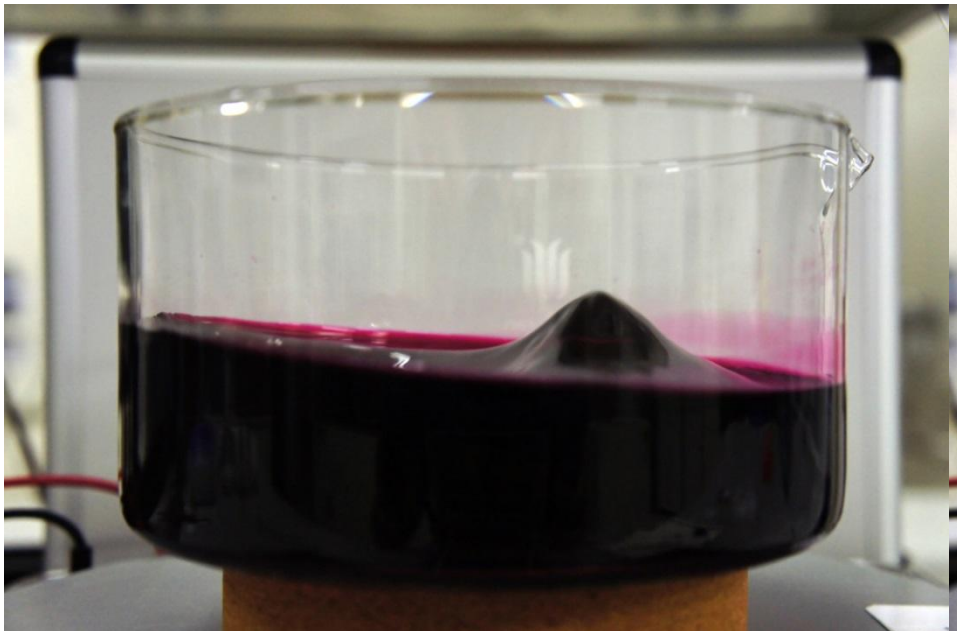


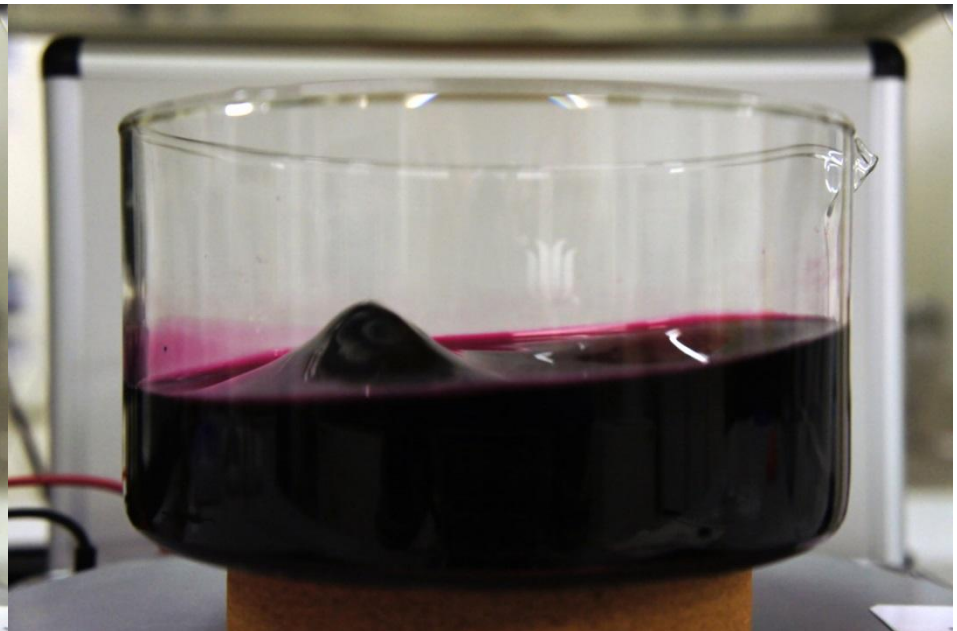
The 2*p* Wave



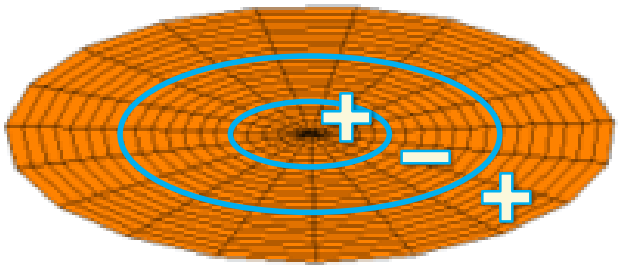
1 nodal line (angular node)

$n = 2,\ l = 1 \implies 2p$



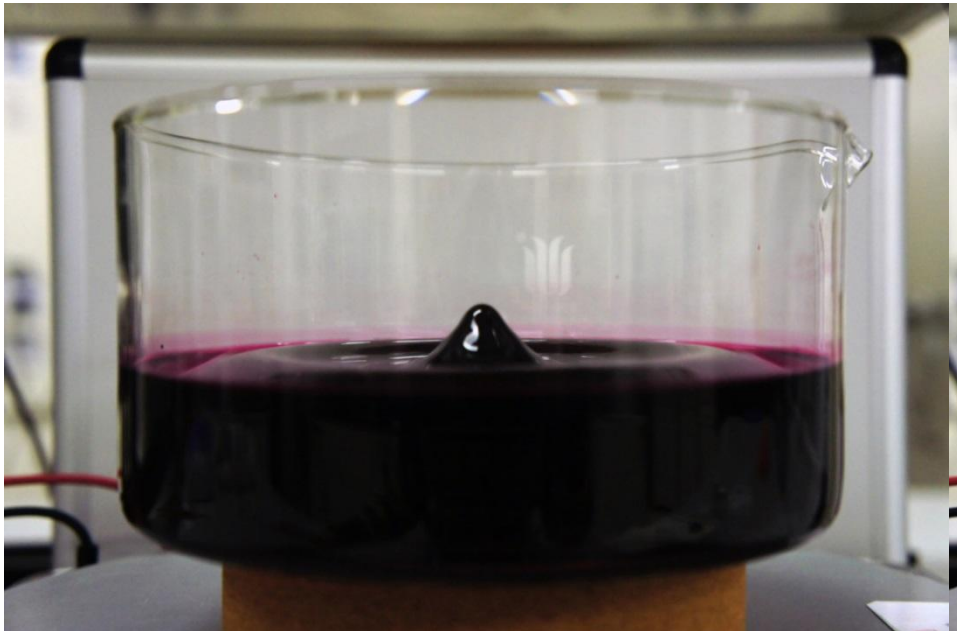


The 3*s* Wave

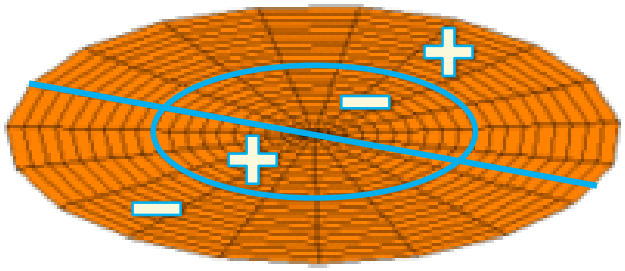


2 nodal rings (radial nodes)

$n = 3,\ l = 0 \implies 3s$



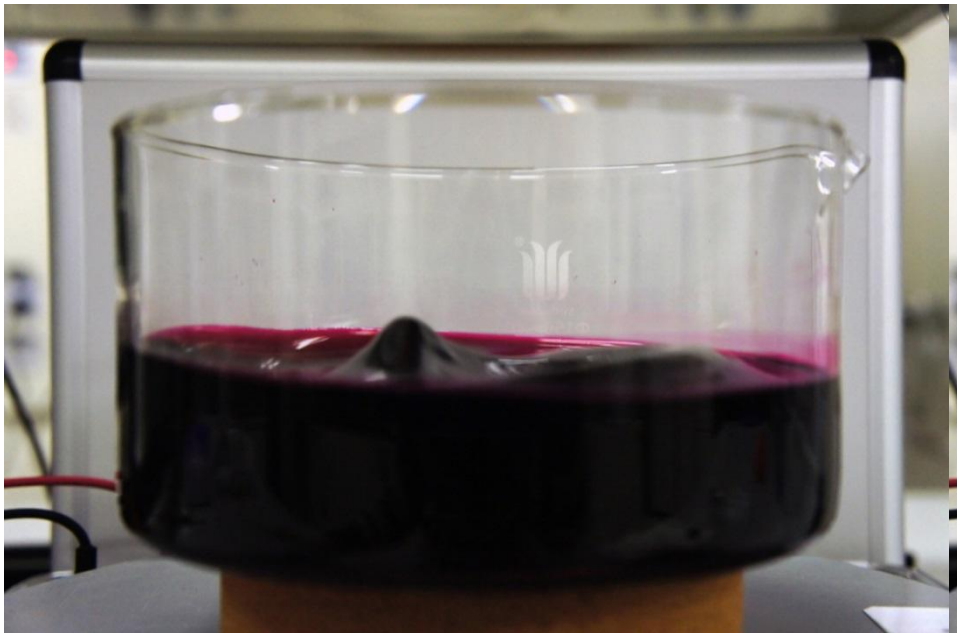
The 3*p* Wave



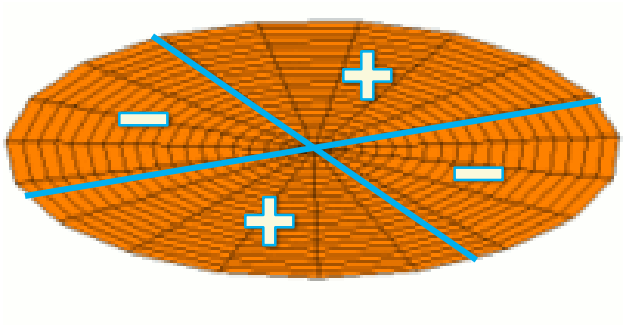
1 nodal ring (radial node)

1 nodal line (angular node)

$n = 3,\ l = 1 \implies 3p$

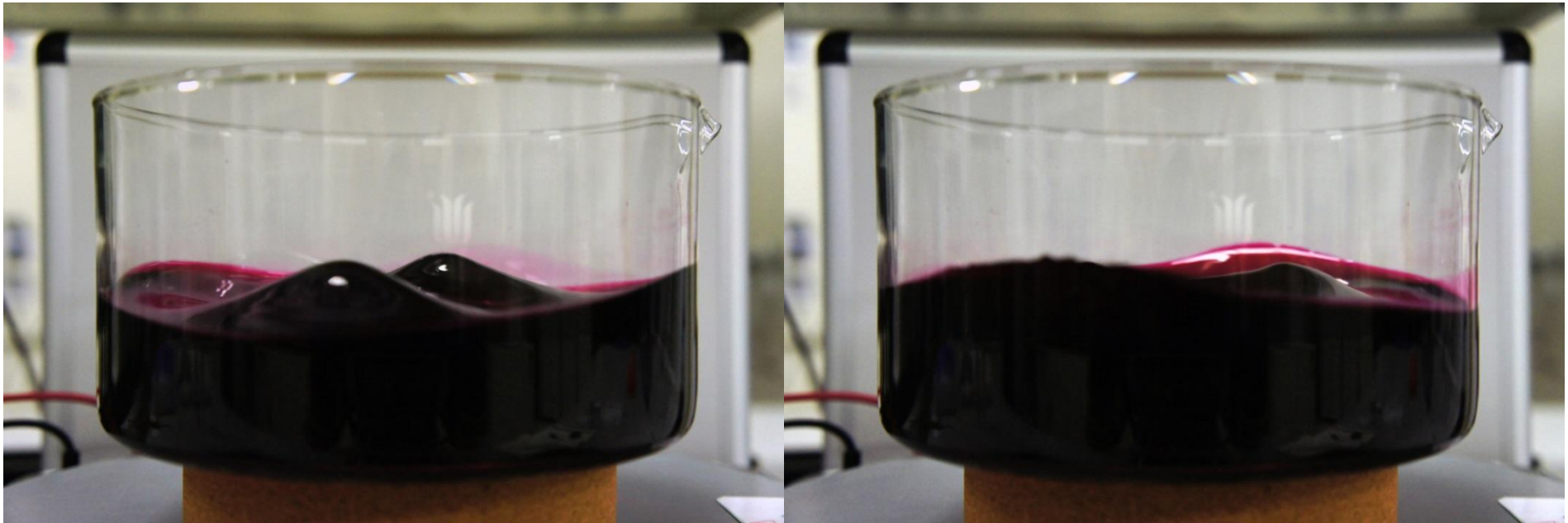


The 3*d* Wave



2 nodal lines (angular nodes)

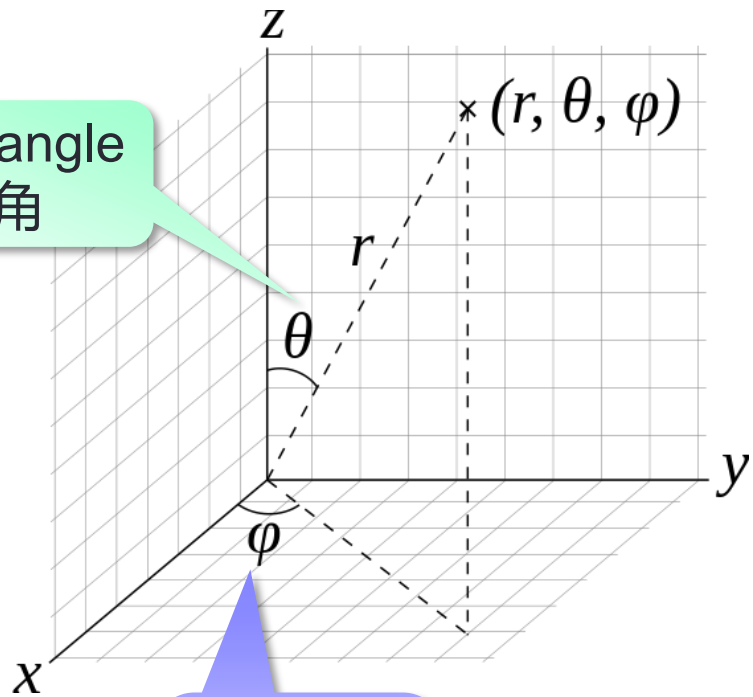
$n = 3,\ l = 2 \Longrightarrow 3d$



Outline

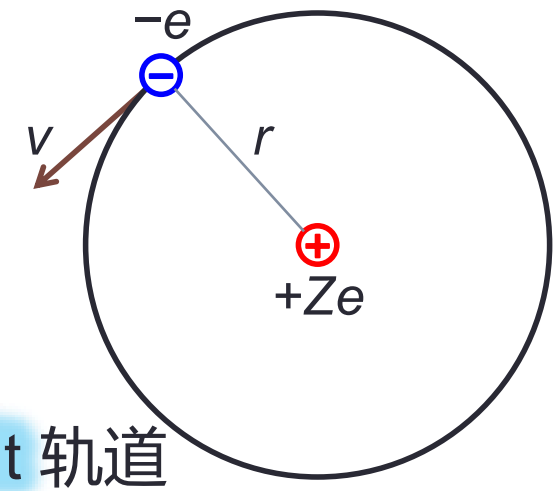
- Oscillations in 2D
- Atomic orbitals: Appearance
- Atomic orbitals: Properties

Spherical Polar Coordinates 球极坐标



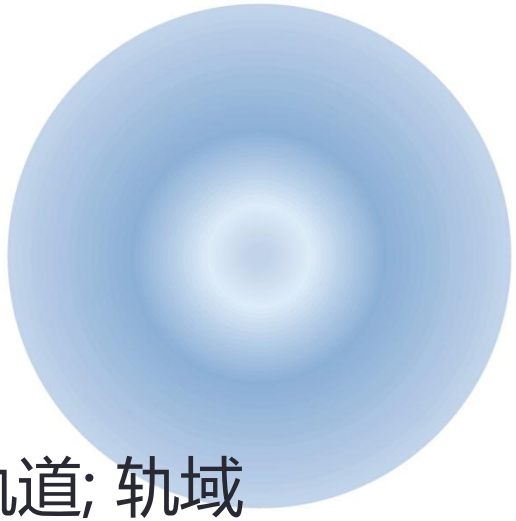
Polar angle
极角

Azimuthal angle
方位角



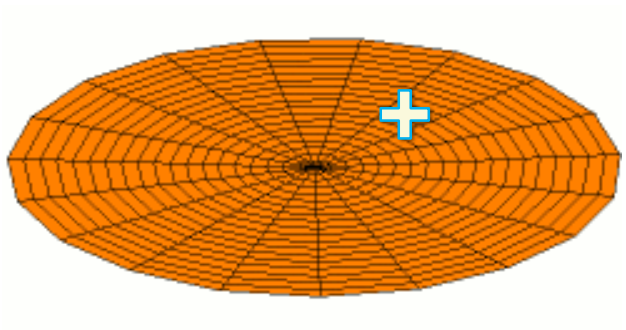
Orbit 轨道

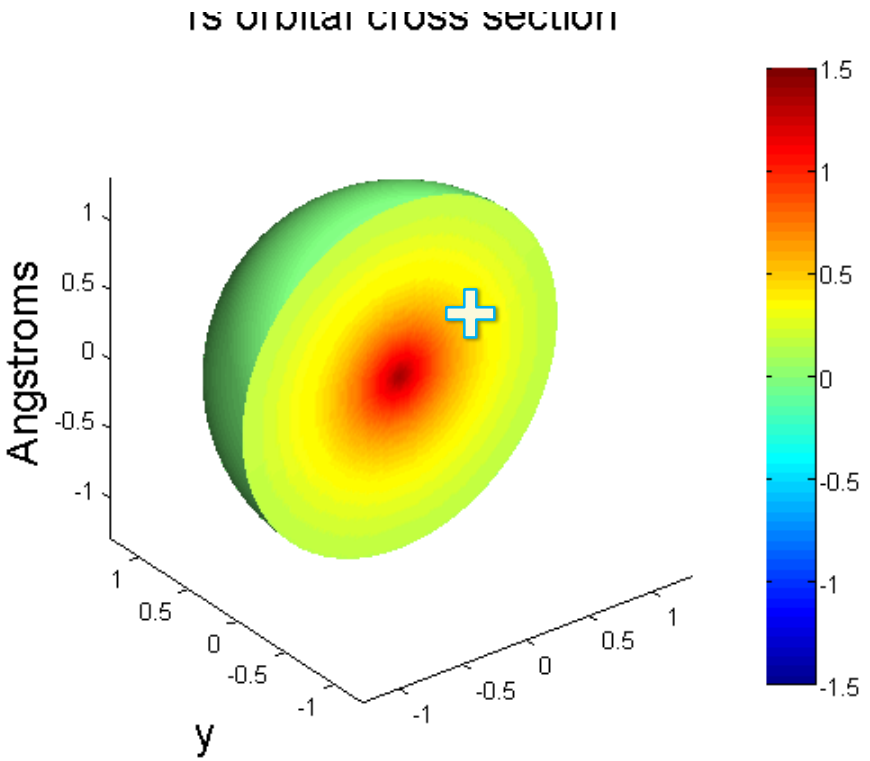
$$E_p = -\frac{1}{4\pi\varepsilon_0}\frac{Ze^2}{r}$$



Orbital 轨道; 轨域

The 1*s* Wave in 3D

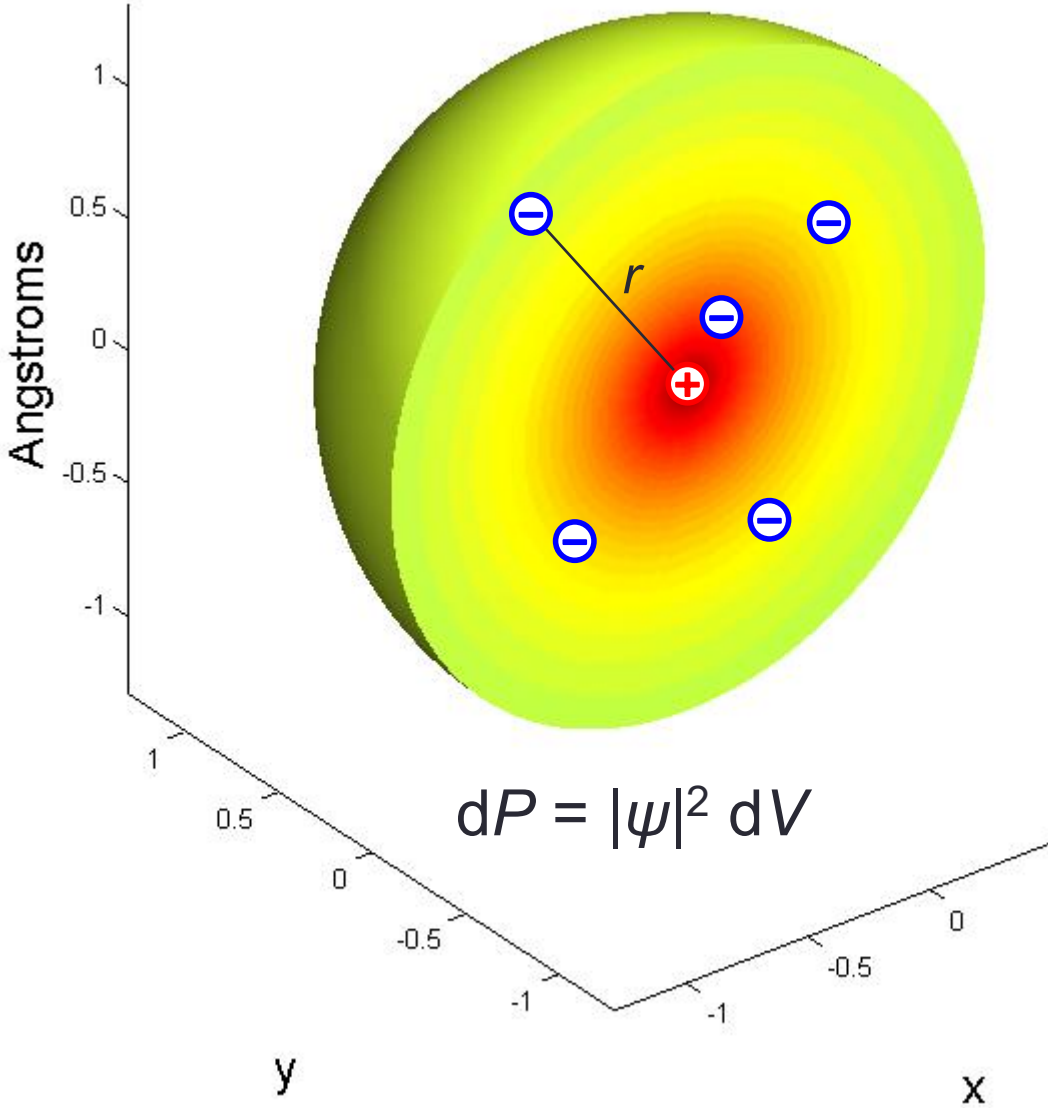




No nodes.

$n = 1,\ l = 0 \Longrightarrow 1s$

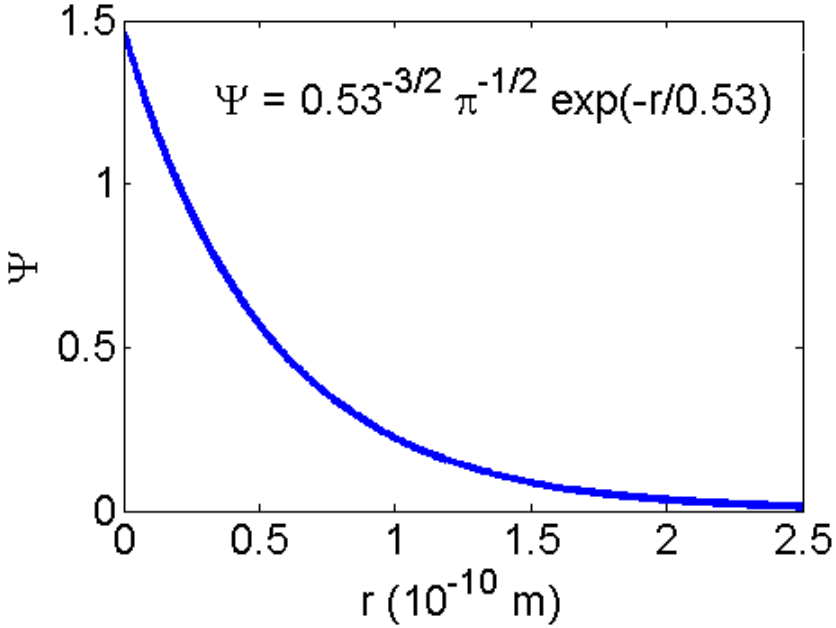
Wave Function of the 1*s* Orbital



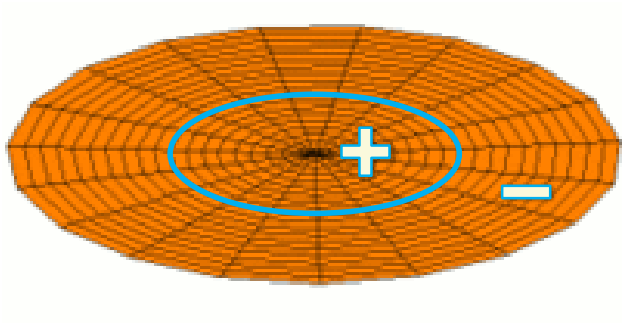
$$\psi_{1s} = \frac{1}{\sqrt{\pi}} a_0^{-3/2} \cdot \exp\left(-\frac{r}{a_0}\right)$$

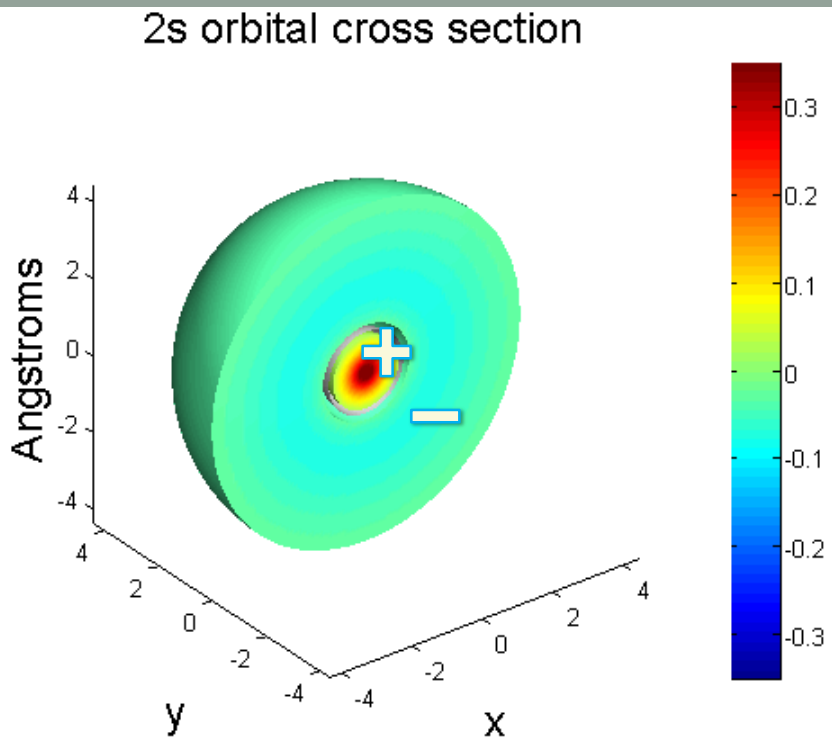
Bohr radius a_0 = 53 pm = 0.53 Å

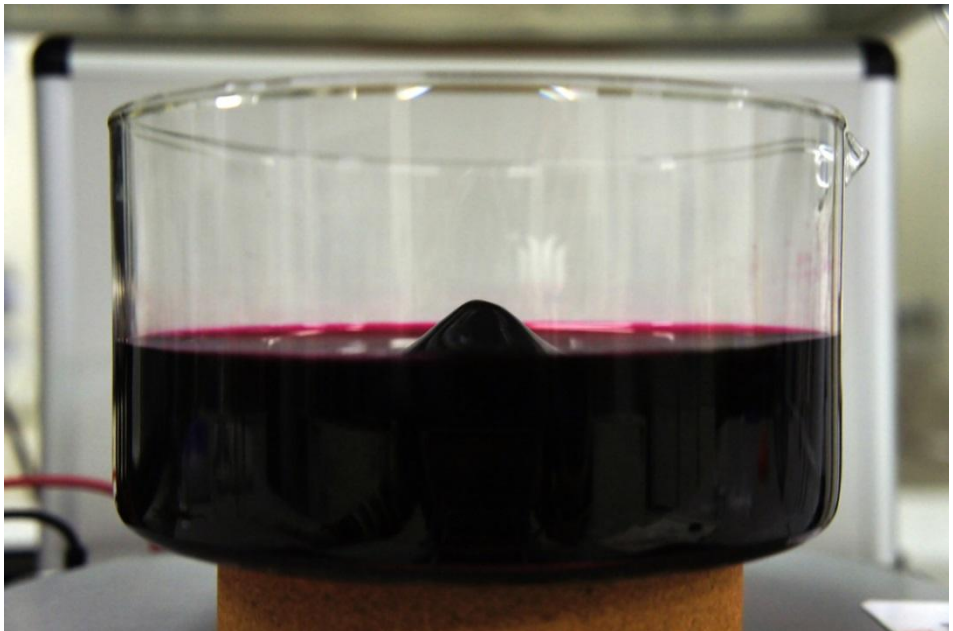
1 Ångstrom = 10^{-10} m



The 2*s* Wave in 3D



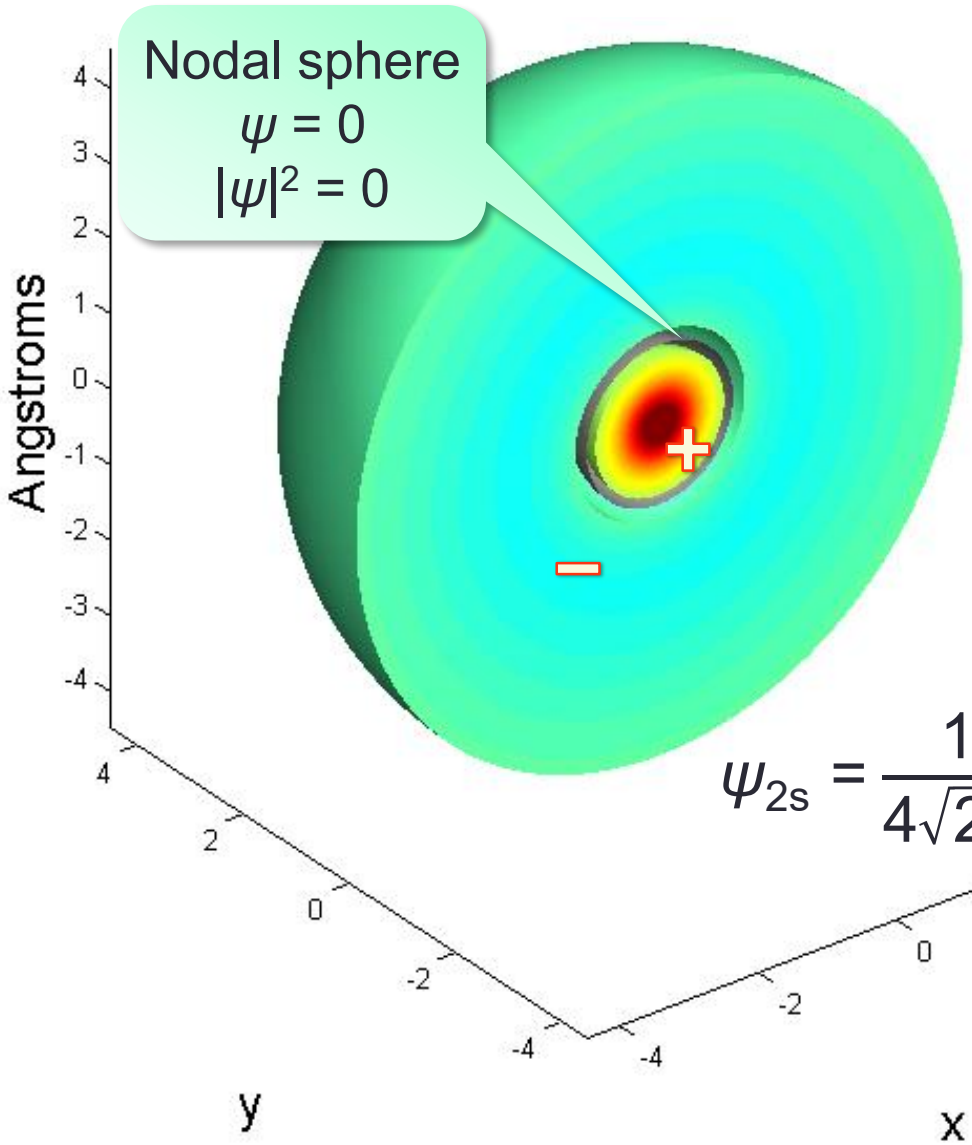


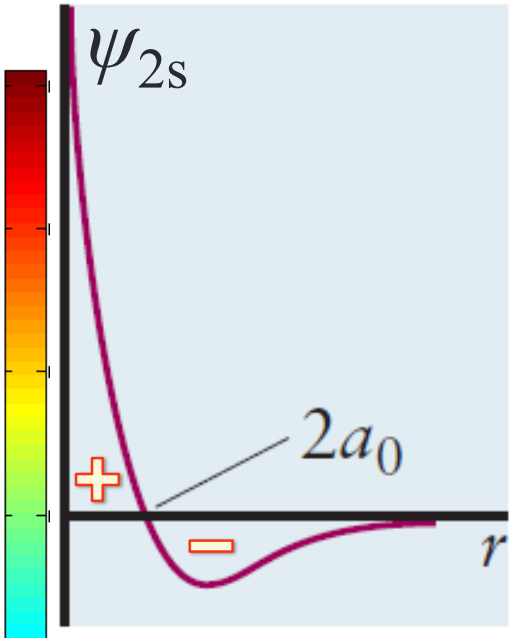


1 nodal sphere (radial node)

$n = 2,\ l = 0 \Longrightarrow 2s$

Wave Function of the 2*s* Orbital

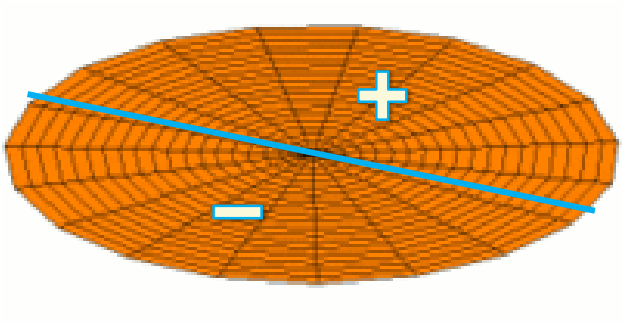


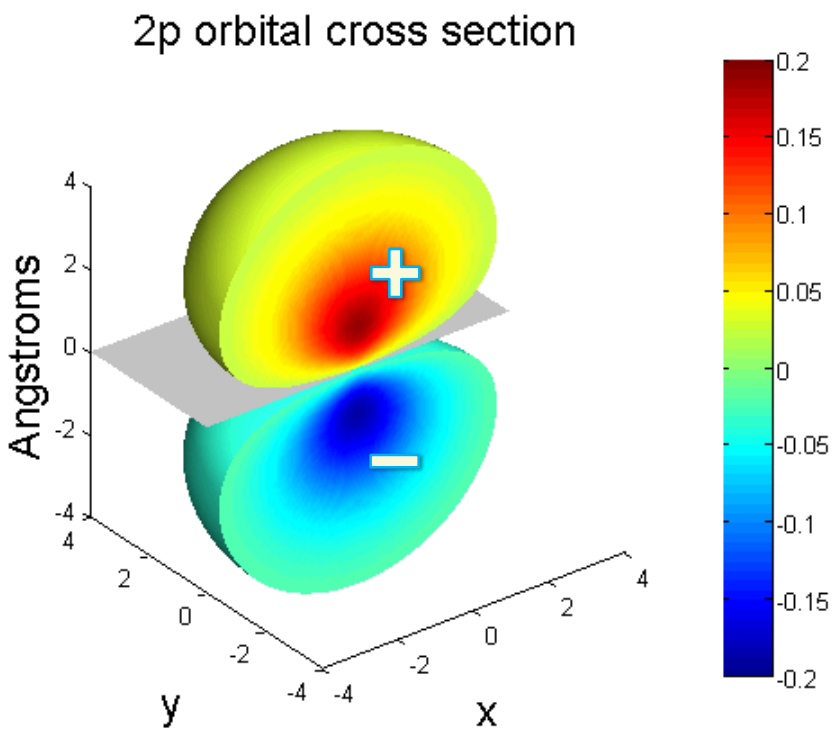


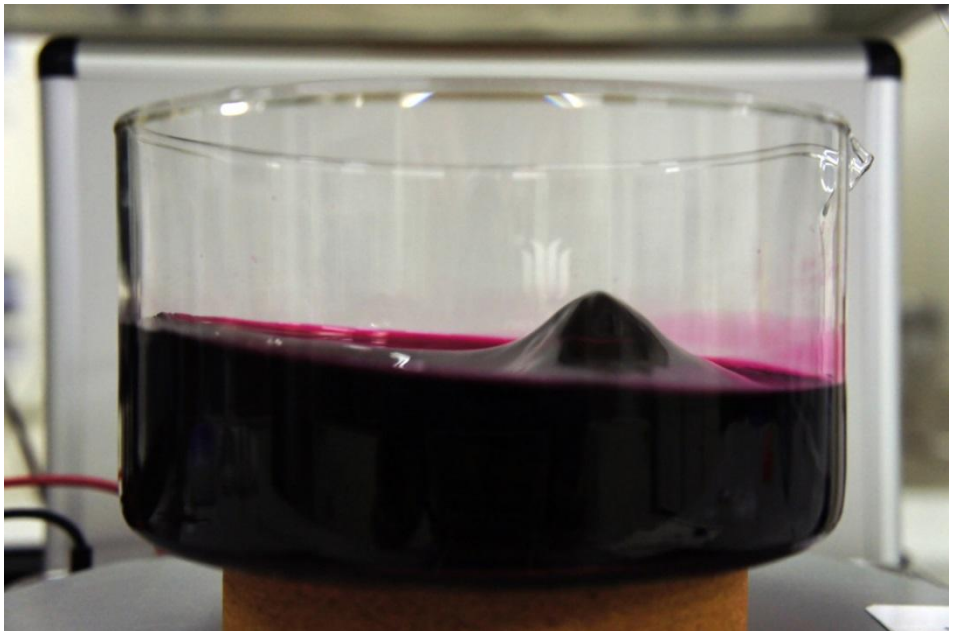
$$\psi_{2s} = \frac{1}{4\sqrt{2\pi}} a_0^{-3/2} \cdot \left(2 - \frac{r}{a_0}\right) \exp\left(-\frac{r}{2a_0}\right)$$

OGB8 p.182

The 2*p* Wave in 3D





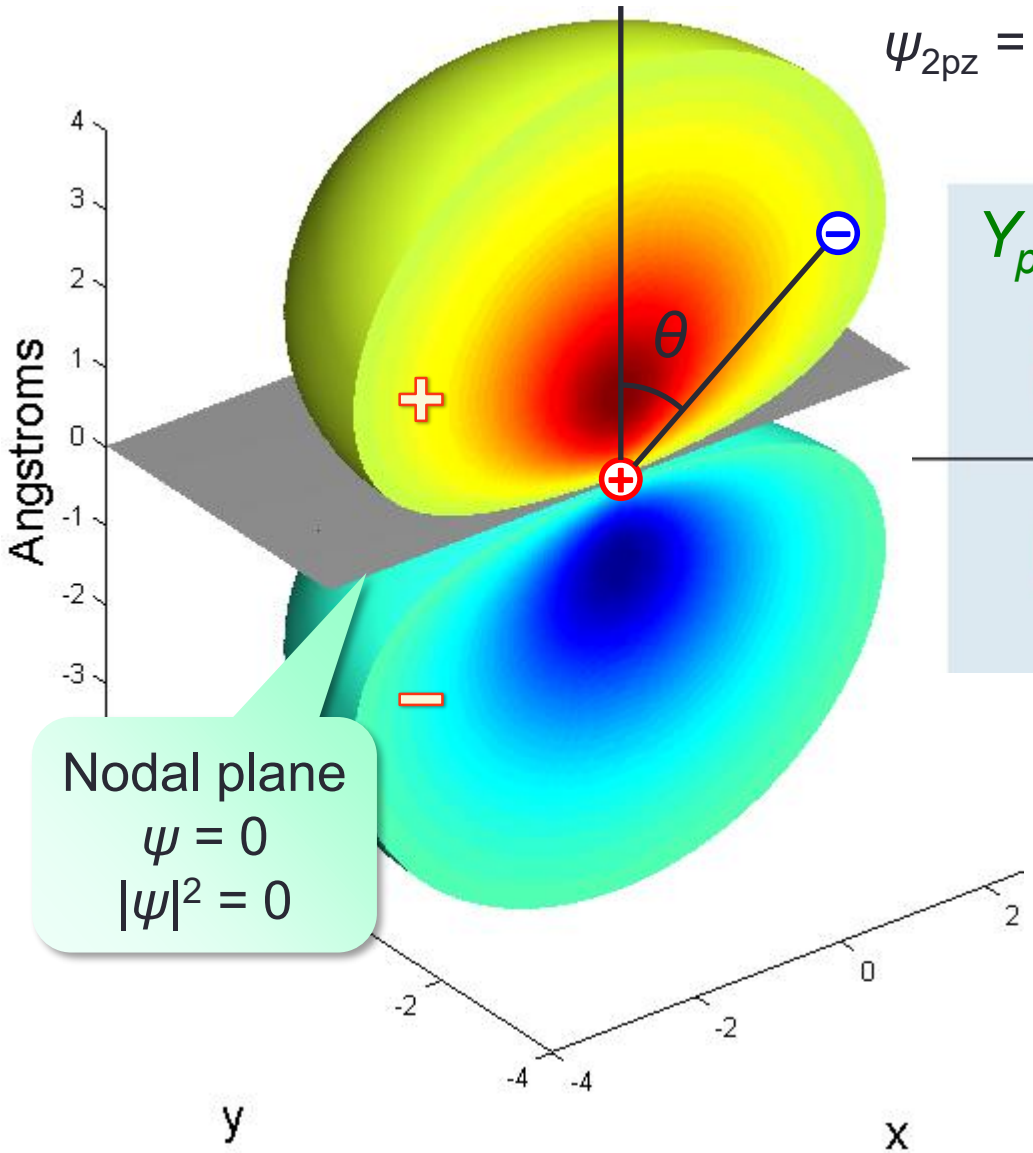


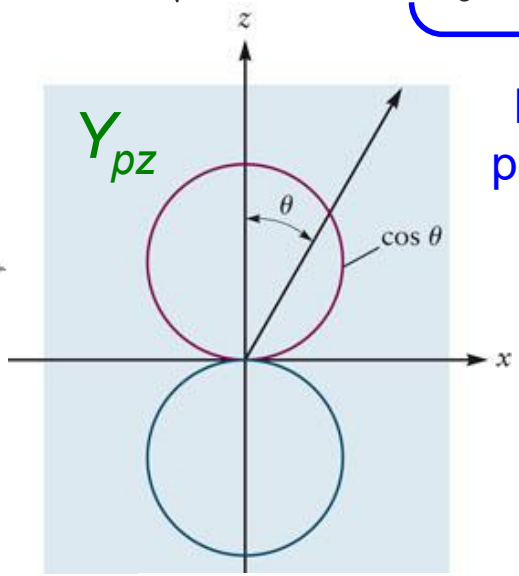
1 nodal plane (angular node)

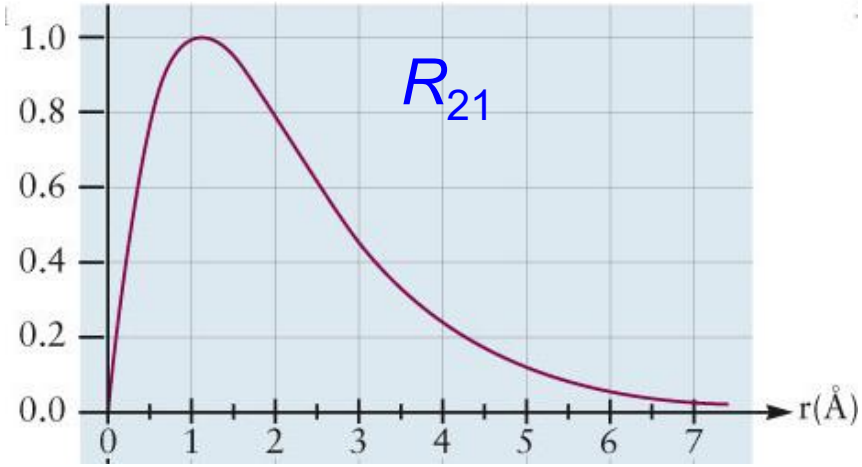
$n = 2,\ l = 1 \Rightarrow 2p$

Wave Function of the $2p_z$ Orbital

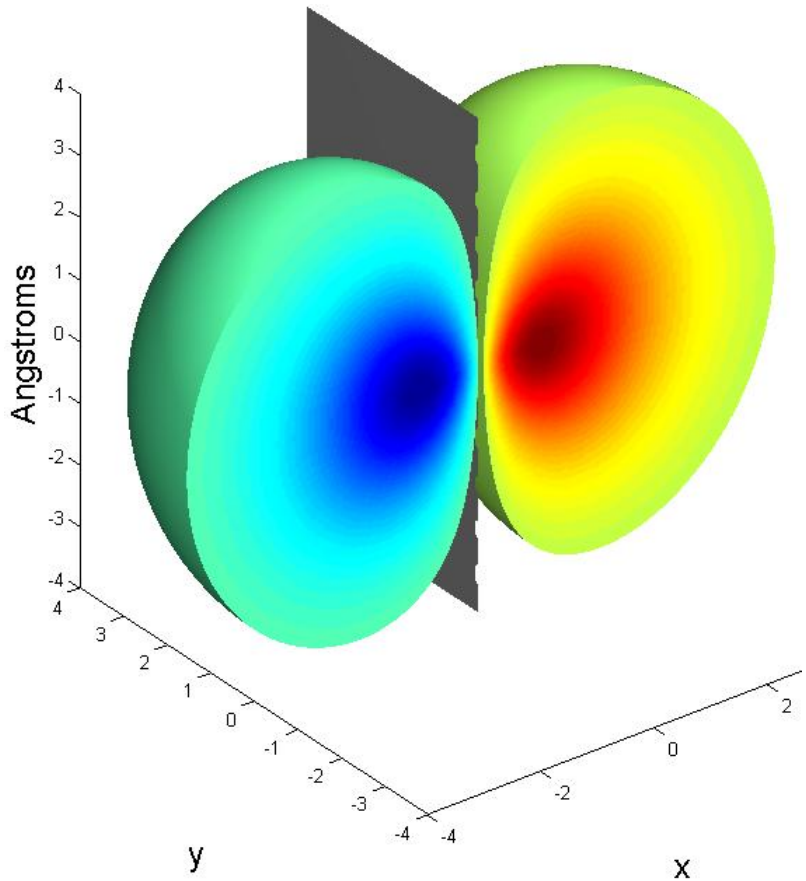
$$\psi_{2pz} = \frac{1}{4\sqrt{2\pi}} a_0^{-3/2} \cdot \underbrace{\frac{r}{a_0} \exp\left(-\frac{r}{2a_0}\right)}_{\text{Radial part, } R_{21}} \cdot \underbrace{\cos\theta}_{\text{Augular part, } Y_{pz}}$$



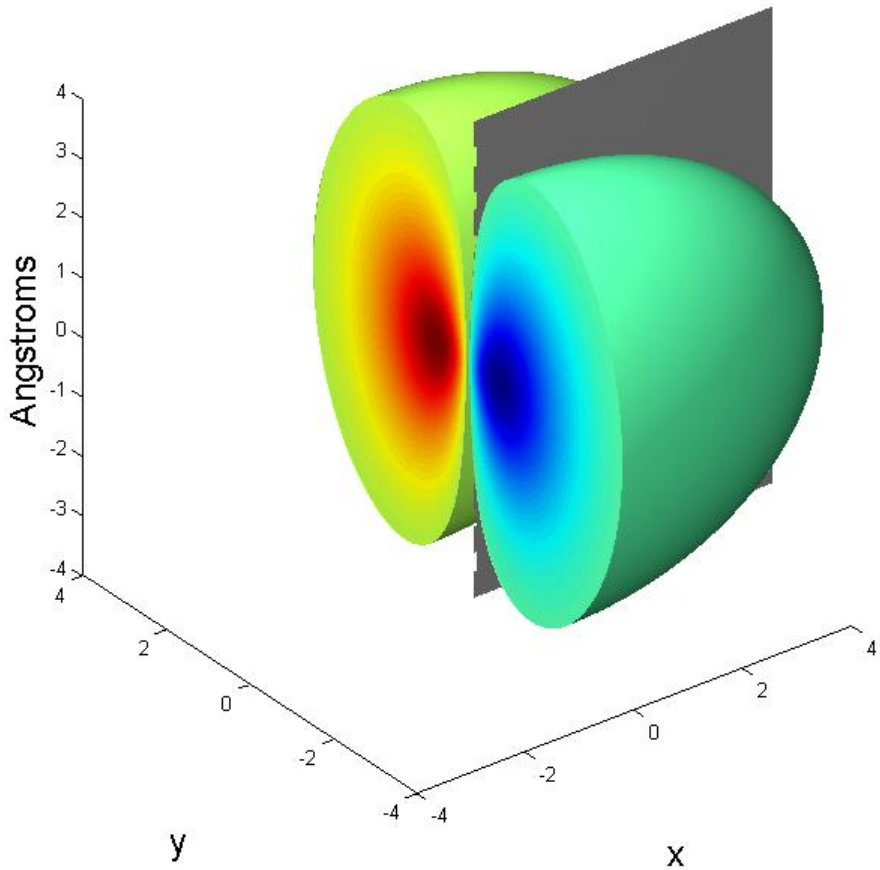




The $2p_x$ and $2p_y$ Orbitals

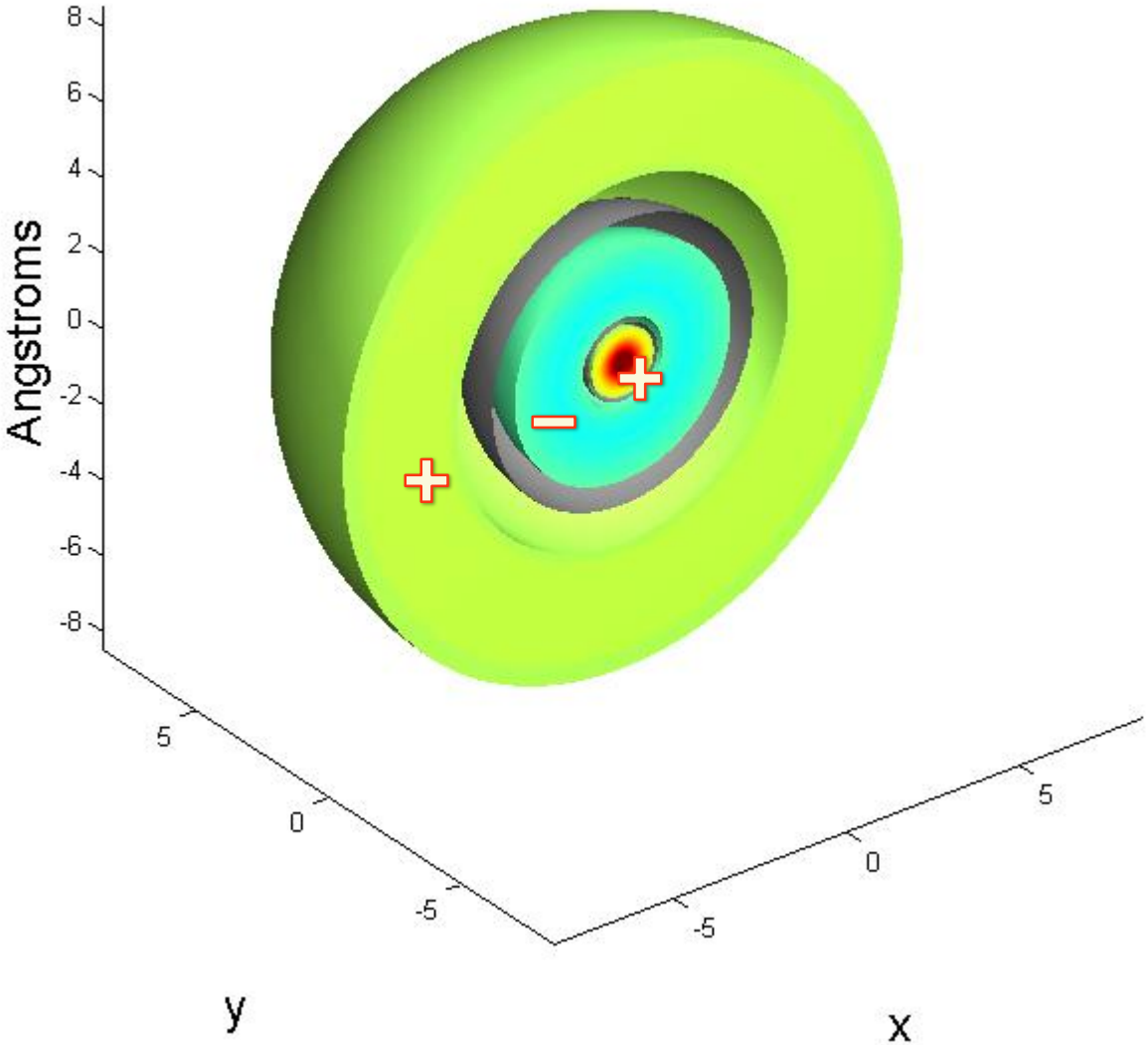


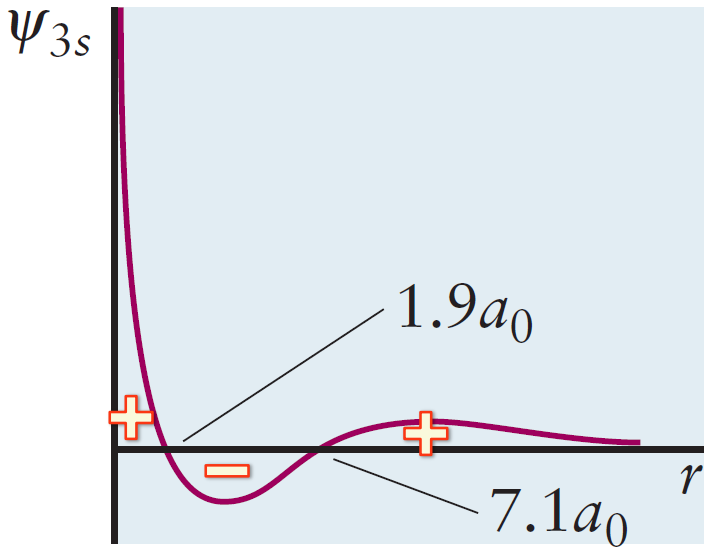
$2p_x$



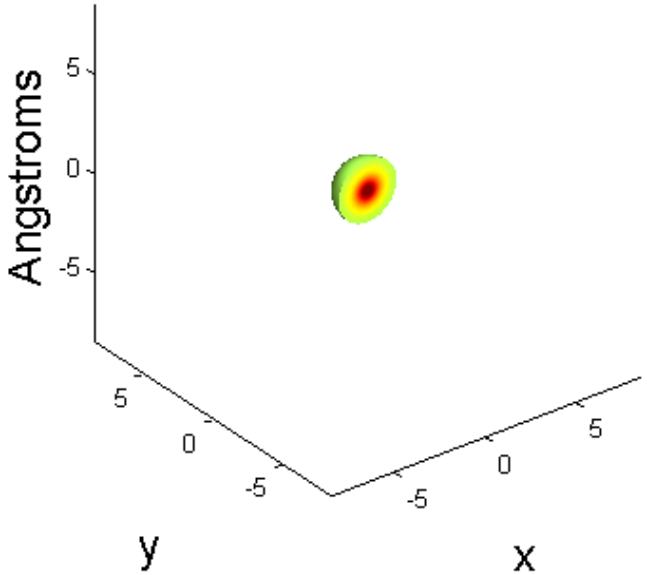
$2p_y$

The 3*s* Orbital



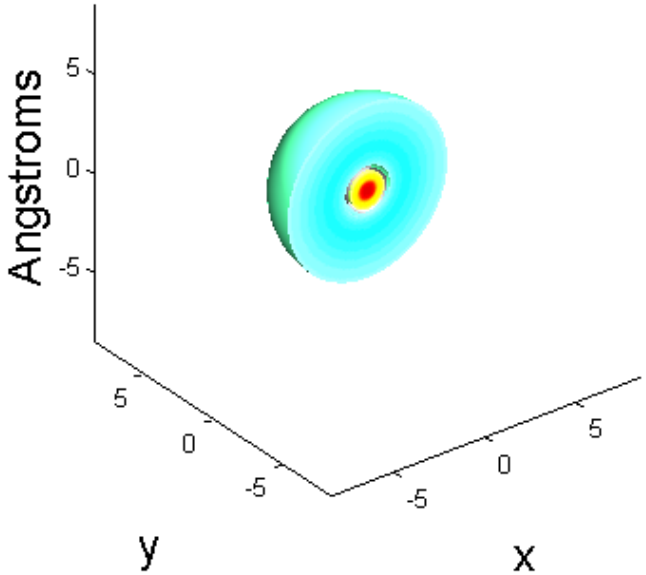


Comparison of 1*s*, 2*s*, 3*s* Orbitals



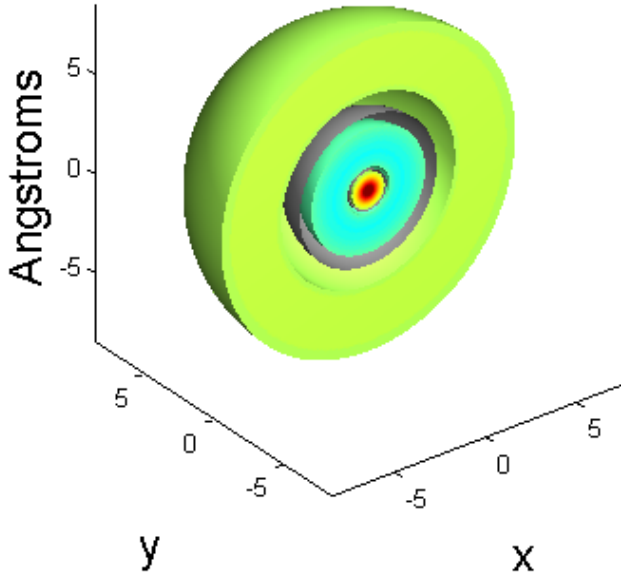
1*s*

n = 1, *l* = 0.



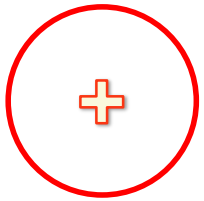
2*s*

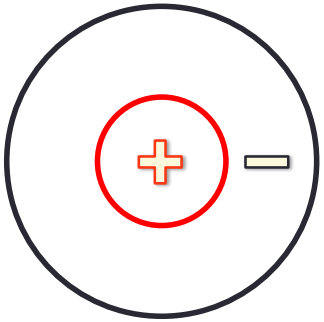
n = 2, *l* = 0.

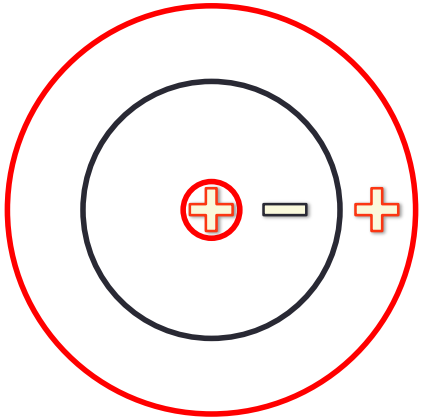


3*s*

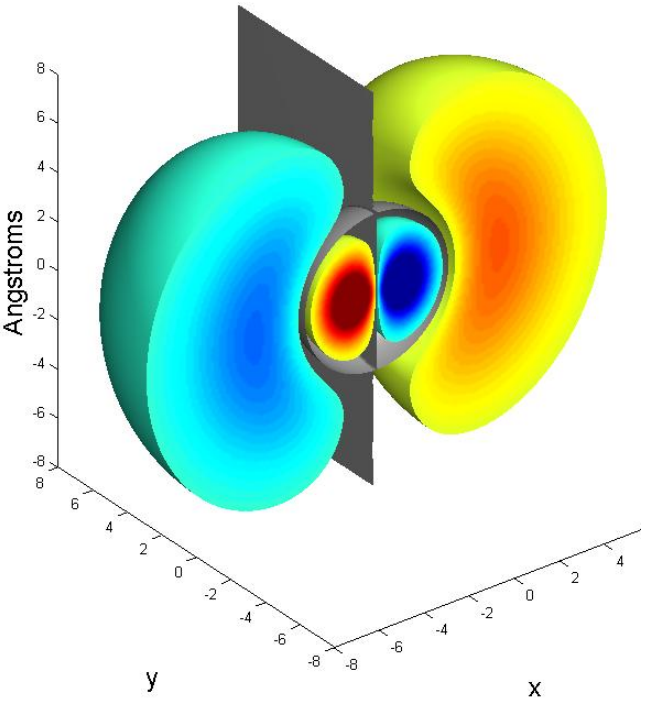
n = 3, *l* = 0.



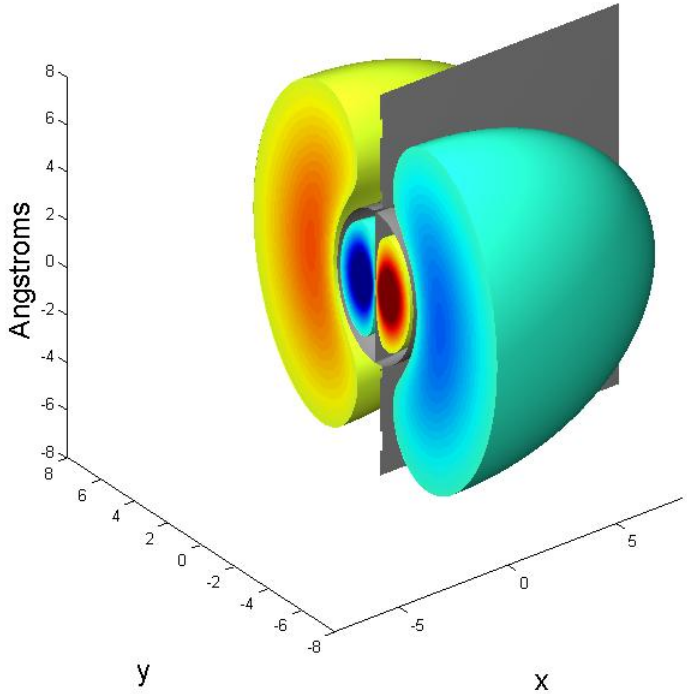




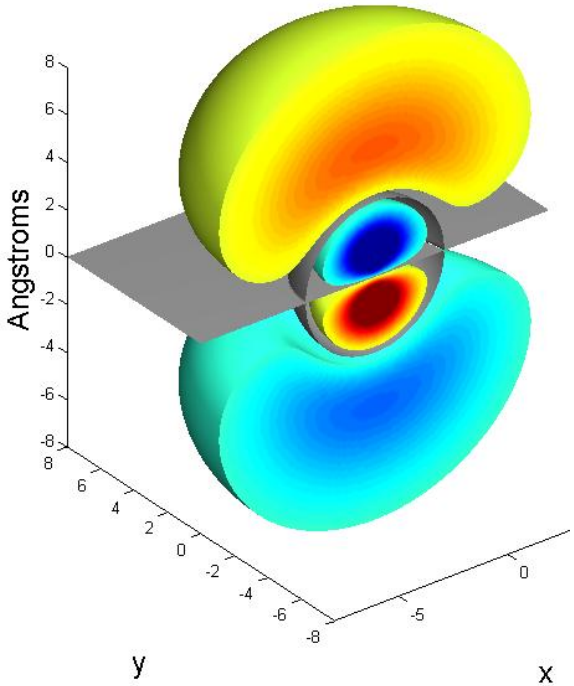
The 3*p* Orbitals



$3p_x$

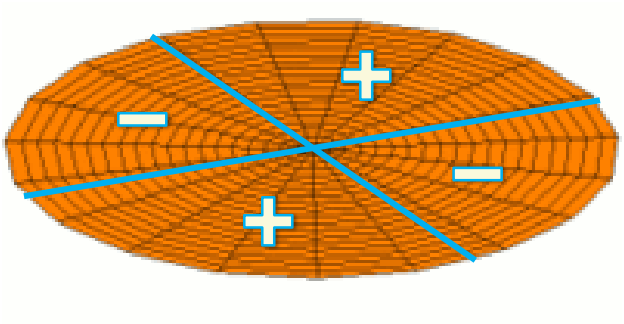


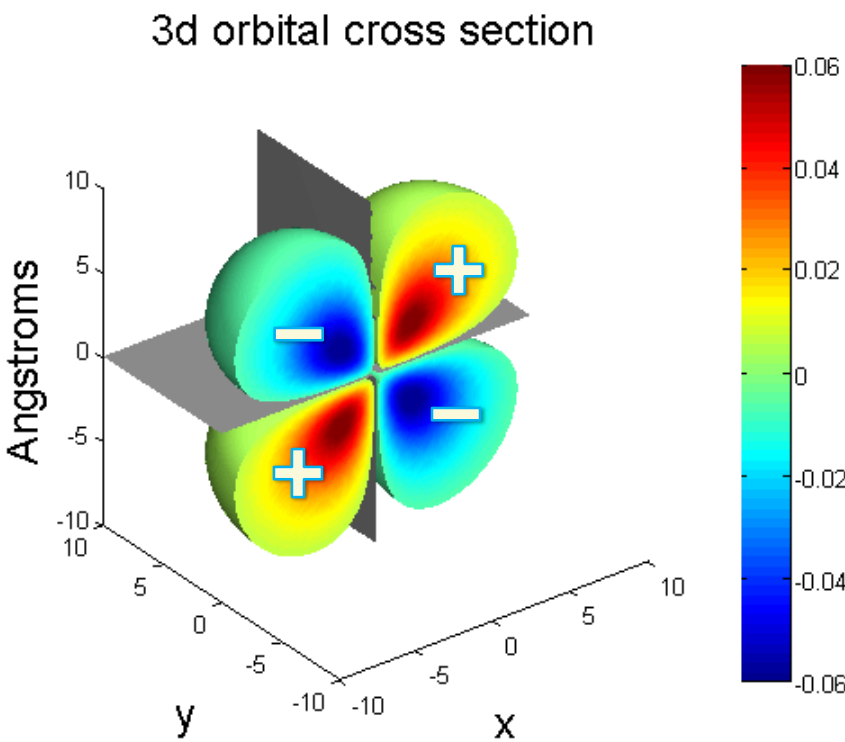
$3p_y$



$3p_z$

The 3*d* Wave in 3D

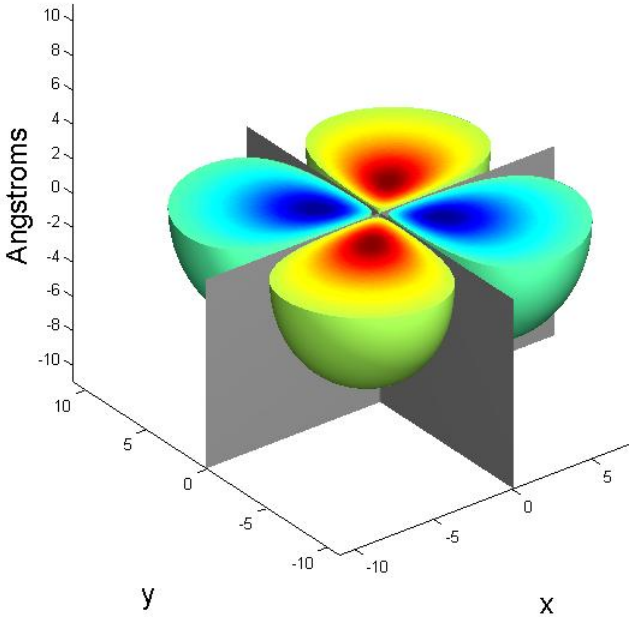




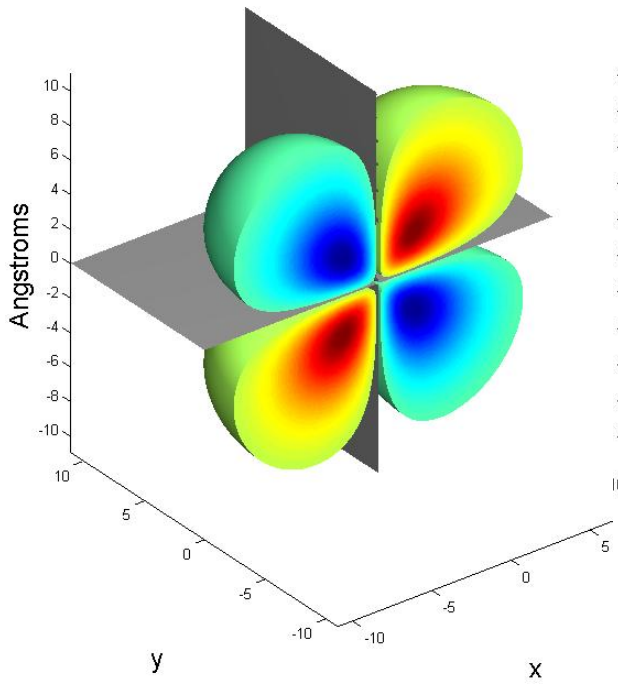
2 nodal planes (angular nodes)

$n = 3,\ l = 2 \implies 3d$

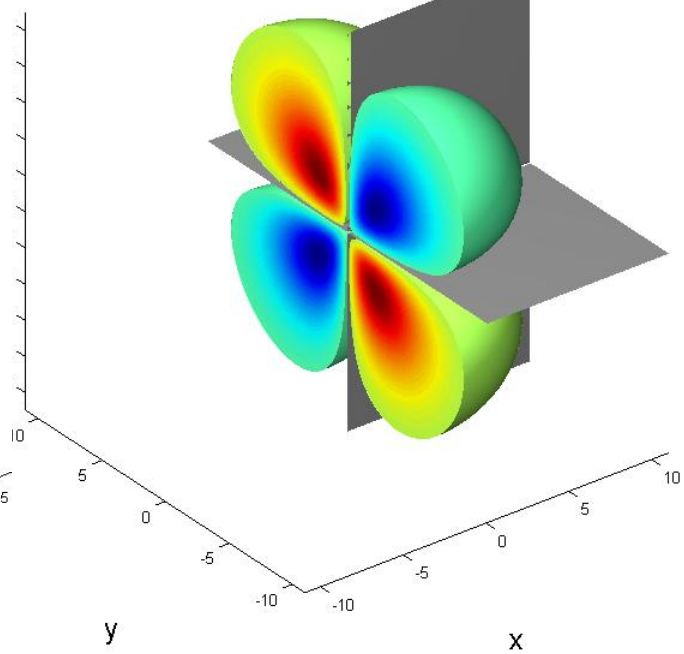
The 3*d* Orbitals (1)



$3d_{xy}$

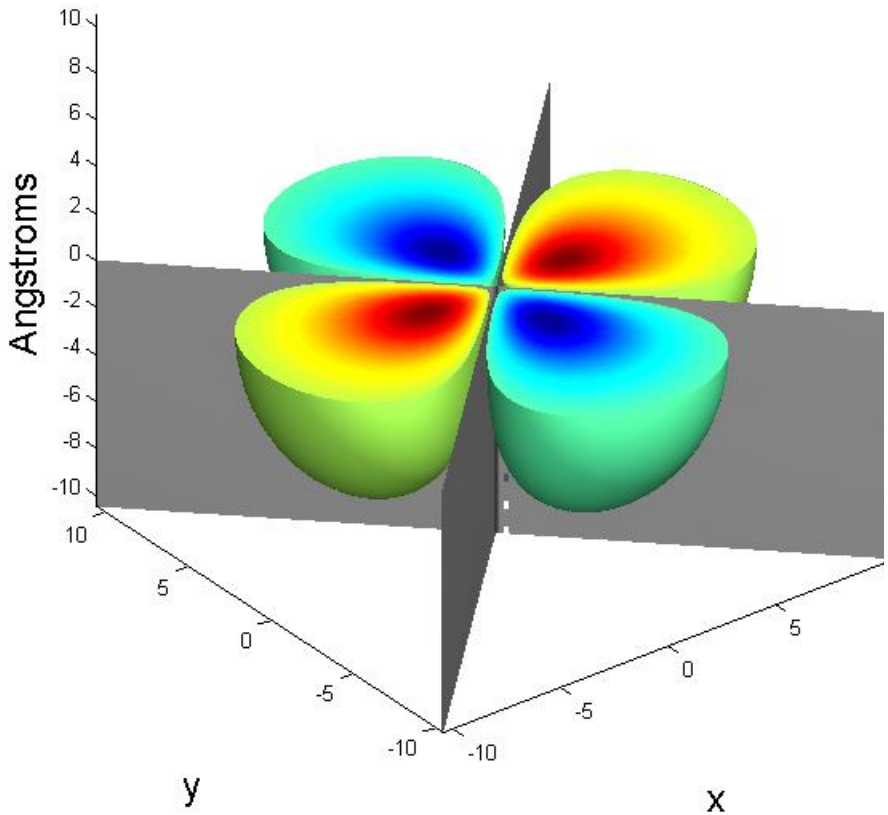


$3d_{xz}$

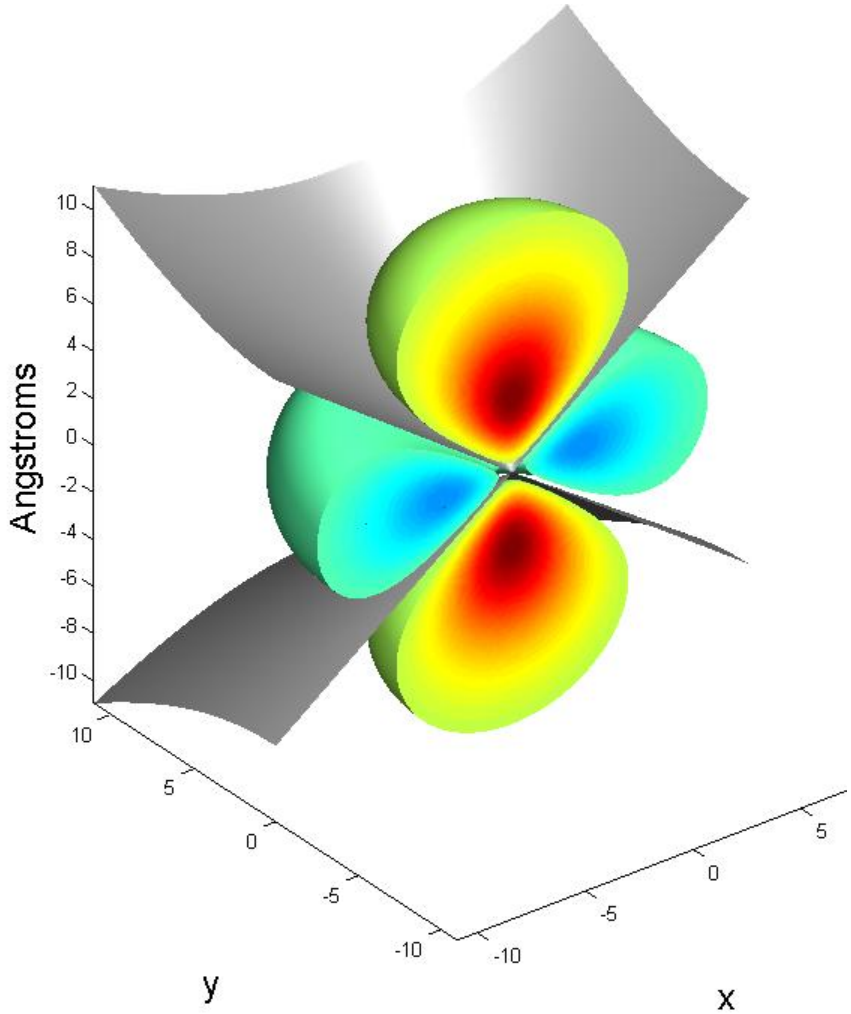


$3d_{yz}$

The 3*d* Orbitals (2)

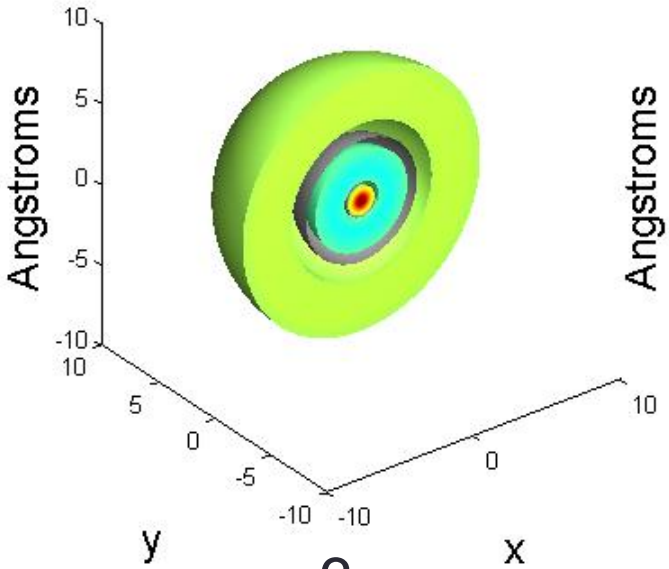


$3d_{x2-y2}$



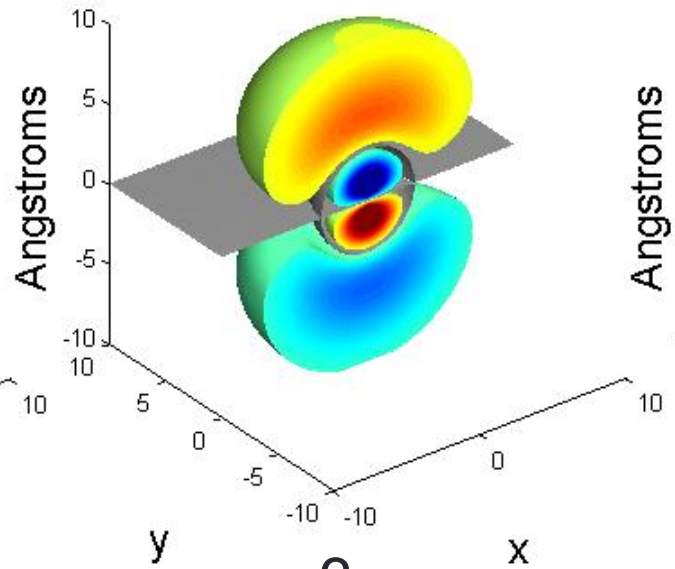
$3d_{z2}$

Comparison of 3*s*, 3*p*, 3*d* Orbitals



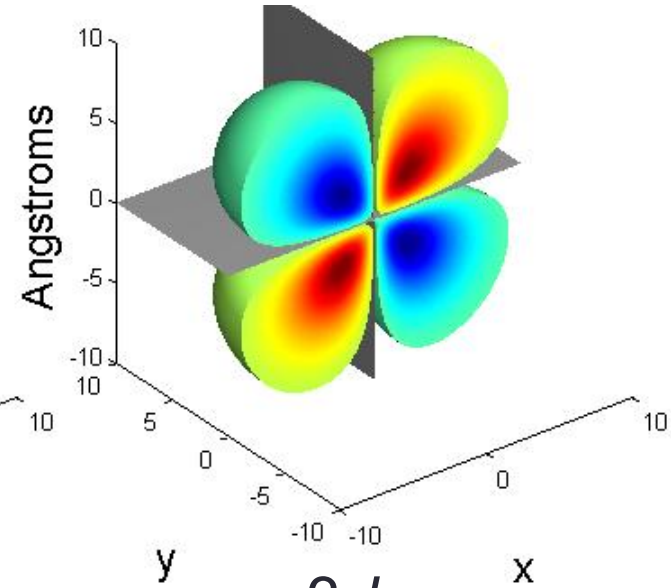
3*s*

n = 3, l = 0.



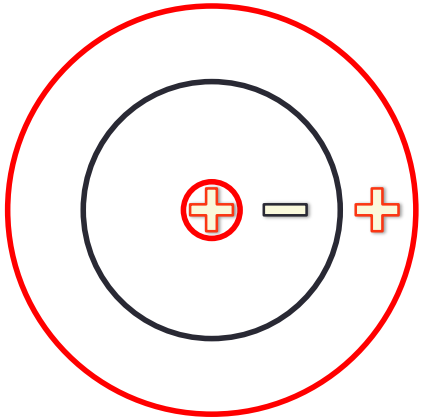
$3p_z$

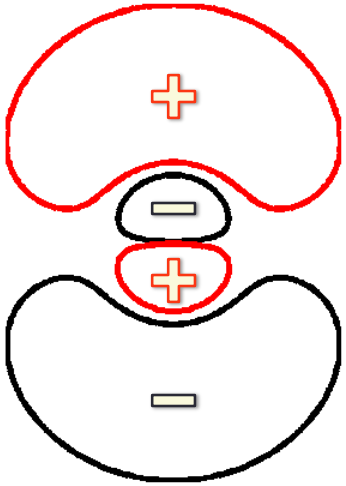
n = 3, l = 1.

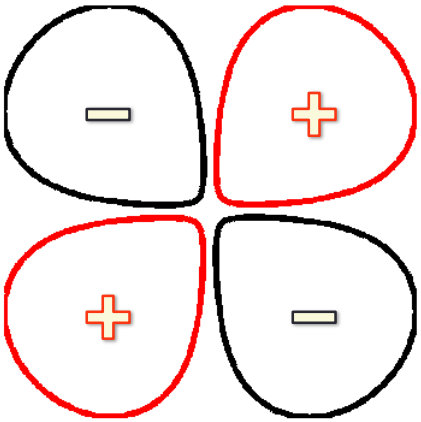


$3d_{xz}$

n = 3, l = 2.







Outline

- Oscillations in 2D
- Atomic orbitals: Appearance
- Atomic orbitals: Properties

Spherical Shell

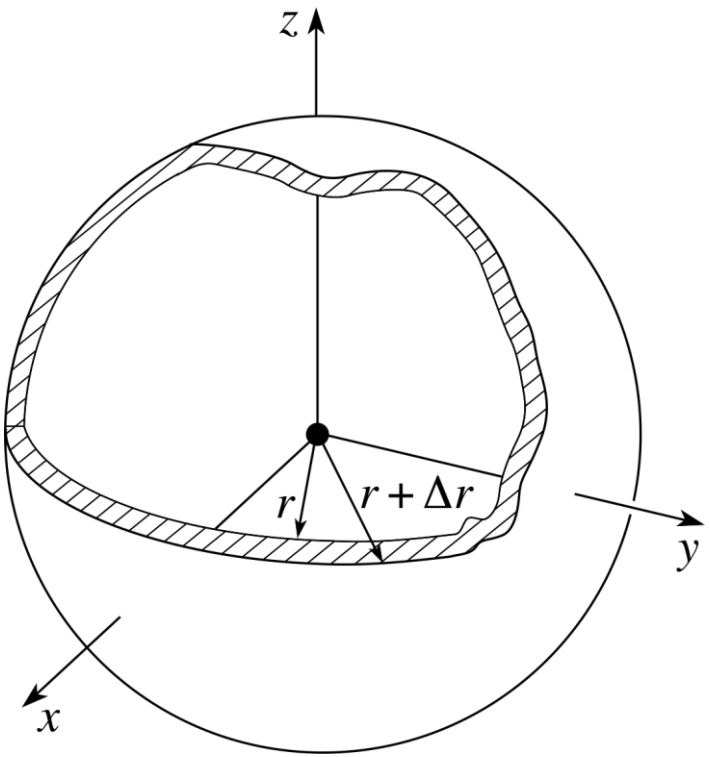


TABLE 5.2

Angular and Radial Parts of Wave Functions for One-Electron Atoms

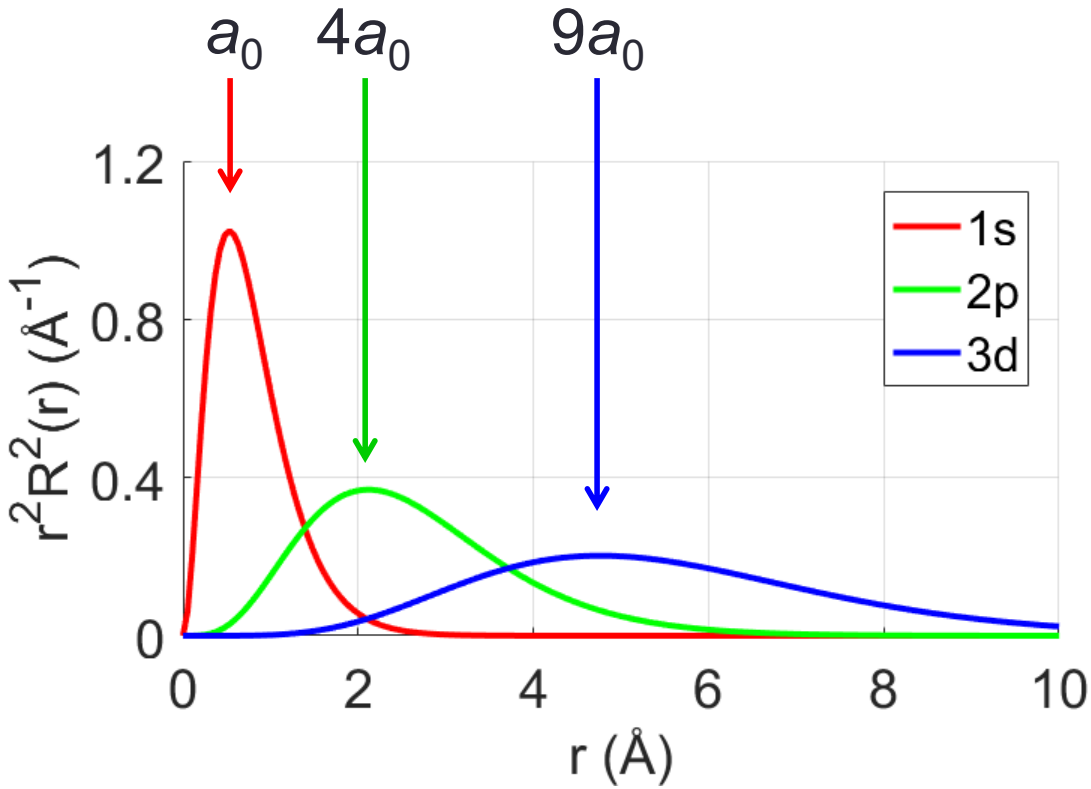
Angular Part $Y(\theta, \phi)$	Radial Part $R_{n\ell}(r)$
$\ell = 0 \quad Y_s = \left(\frac{1}{4\pi}\right)^{1/2}$	$R_{1s} = 2\left(\frac{Z}{a_0}\right)^{3/2} \exp(-\sigma)$
	$R_{2s} = \frac{1}{2\sqrt{2}}\left(\frac{Z}{a_0}\right)^{3/2} (2-\sigma)\exp(-\sigma/2)$
	$R_{3s} = \frac{2}{81\sqrt{3}}\left(\frac{Z}{a_0}\right)^{3/2} (27 - 18\sigma + 2\sigma^2)\exp(-\sigma/3)$
$\ell = 1 \quad Y_{p_x} = \left(\frac{3}{4\pi}\right)^{1/2} \sin\theta\cos\phi$	$R_{2p} = \frac{1}{2\sqrt{6}}\left(\frac{Z}{a_0}\right)^{3/2} \sigma\exp(-\sigma/2)$
$Y_{p_y} = \left(\frac{3}{4\pi}\right)^{1/2} \sin\theta\sin\phi$	$R_{3p} = \frac{4}{81\sqrt{6}}\left(\frac{Z}{a_0}\right)^{3/2} (6\sigma - \sigma^2)\exp(-\sigma/3)$
$Y_{p_z} = \left(\frac{3}{4\pi}\right)^{1/2} \cos\theta$	

$$A = 4\pi r^2$$
$$dV = A dr = 4\pi r^2 dr$$

$$dP = |\psi|^2\, dV = R^2 Y^2\, dV$$
$$= R^2 \cdot \frac{1}{4\pi} \cdot 4\pi r^2 dr$$
$$= r^2 R^2\, dr$$

Orbital Radius

Most probable radius r_{mp}



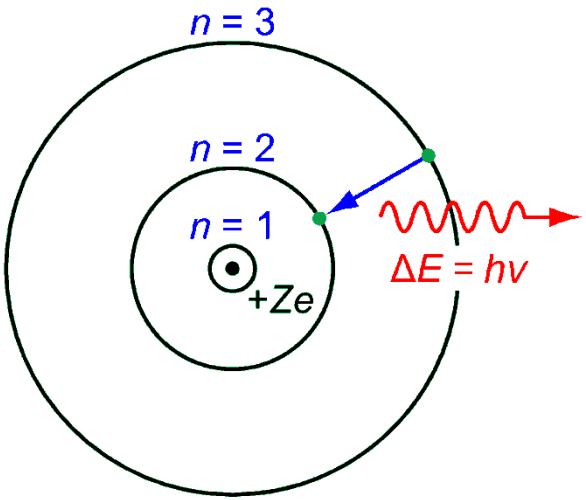
Plot $r^2R^2(r)$ vs. r

For Bohr Model: $r_n = n^2a_0$

For 1*s*, 2*p*, 3*d*, …: $r_{mp,n} = n^2a_0$

Orbital Angular Momentum

$$L = \sqrt{l(l+1)} \cdot h/2\pi \approx l \cdot h/2\pi$$



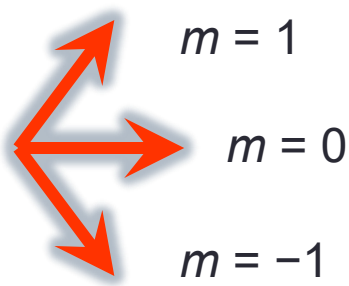
$L_n = nh/2\pi$
in two dimensions

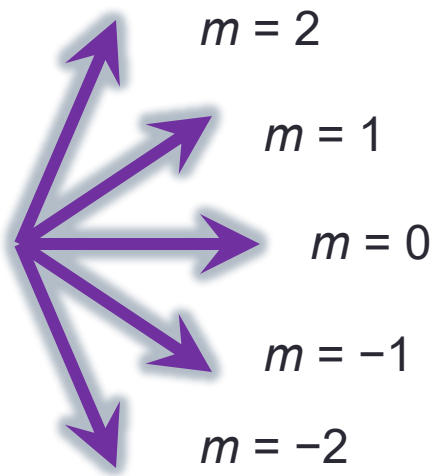
For 1*s*, 2*s* orbitals: $l = 0$, $L = 0$.

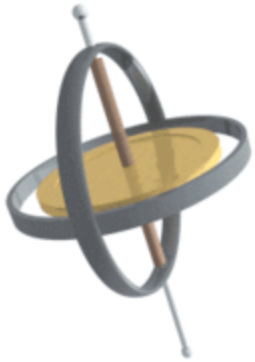
For 2*p*, 3*p* orbitals: $l = 1$, $L = \sqrt{2}\,\hbar$,
L can take 3 orientations.

For 3*d*, 4*d* orbitals: $l = 2$, $L = \sqrt{6}\,\hbar$,
L can take 5 orientations.

磁量子数
Magnetic quantum number *m*







Old *vs*. New Quantum Theory

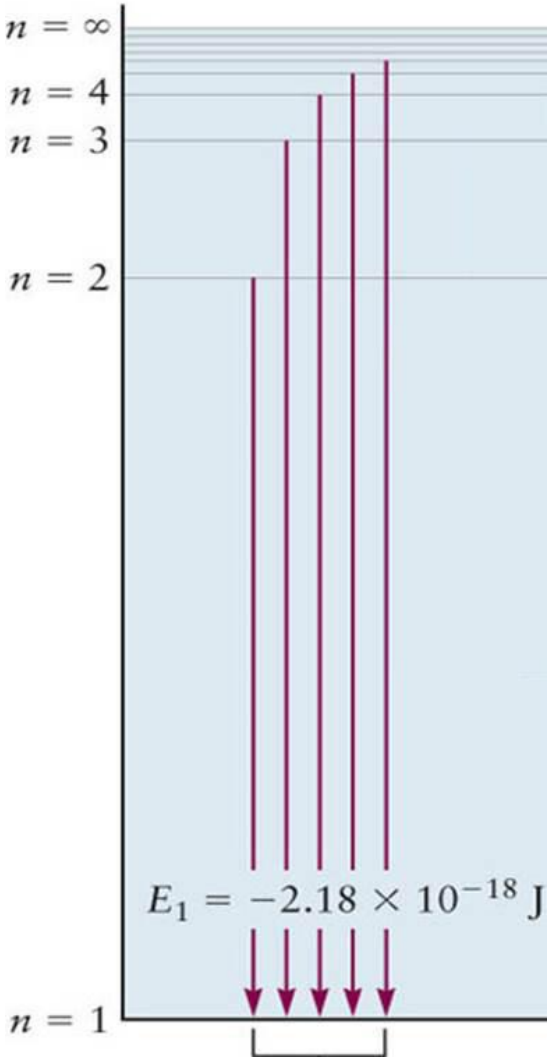
Niels Bohr
(Copenhagen, Cambridge,
1885–1962)

VS

Erwin Schrödinger
(Zürich, 1887–1961)

Orbital Energy

Bohr, OGB8 p.143

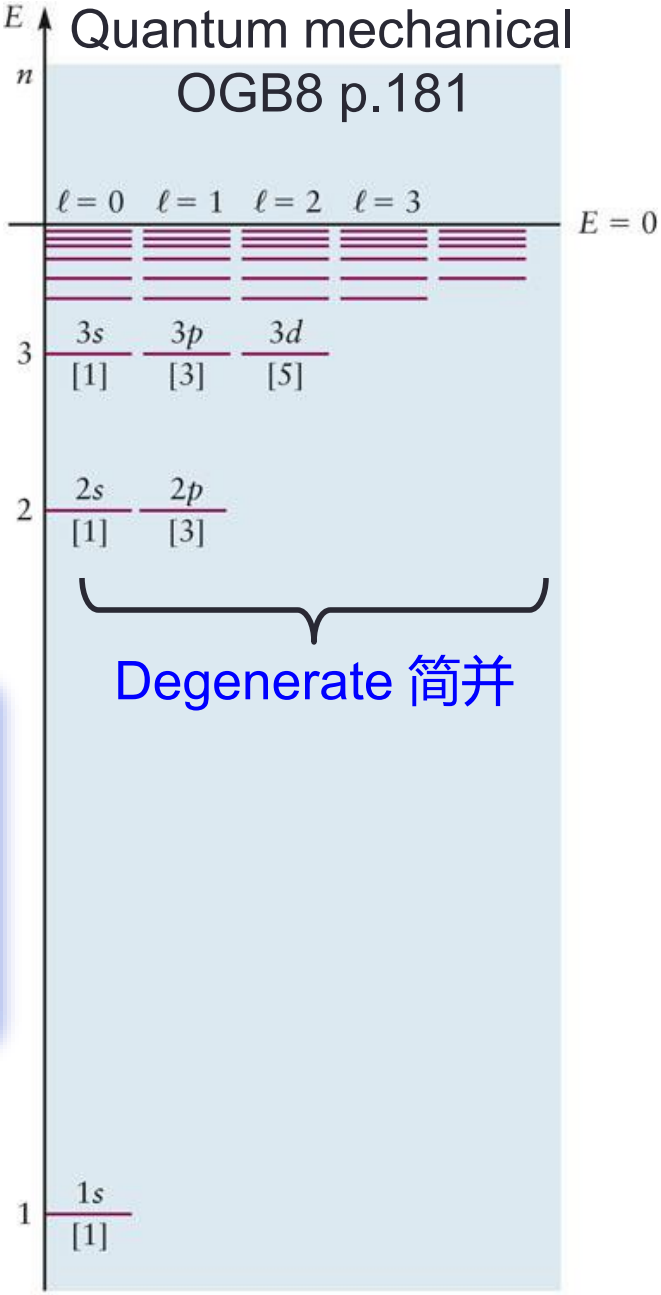


$$E_n = -\frac{\mathrm{Ry}}{n^2},$$

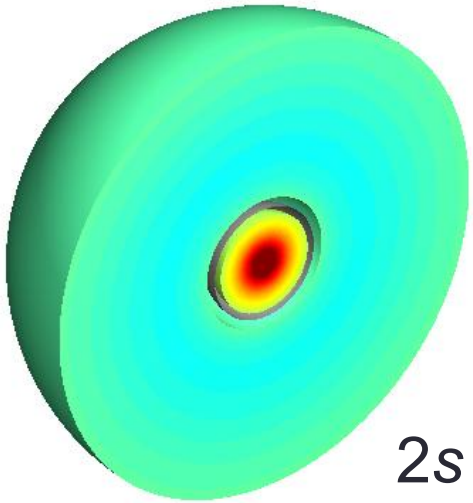
$$\mathrm{Ry} = 2.18\times10^{-18}\ \mathrm{J}$$

$$= 13.6\ \mathrm{eV}$$

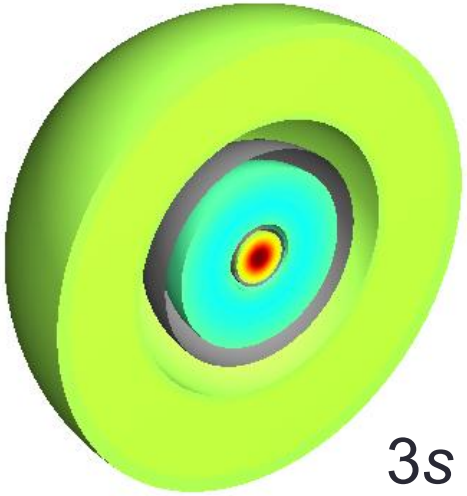
Quantum mechanical
OGB8 p.181



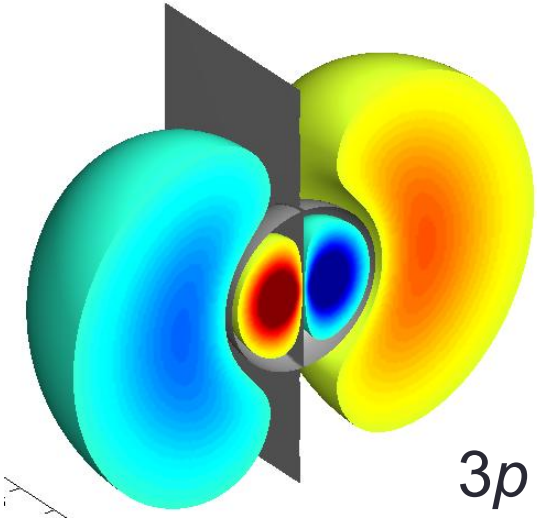
Non-Bohr Orbitals



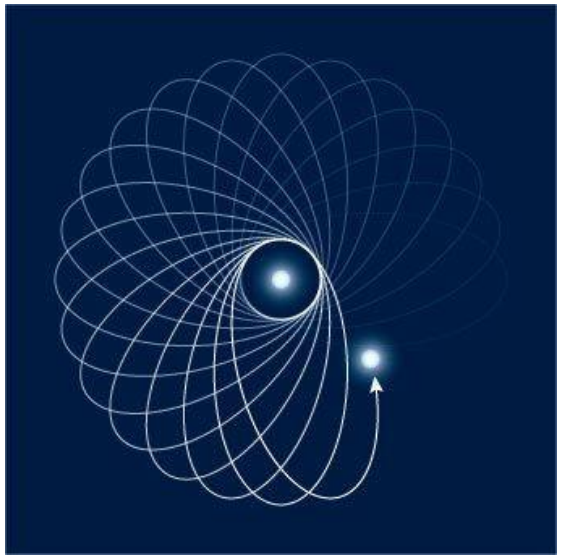
2*s*



3*s*



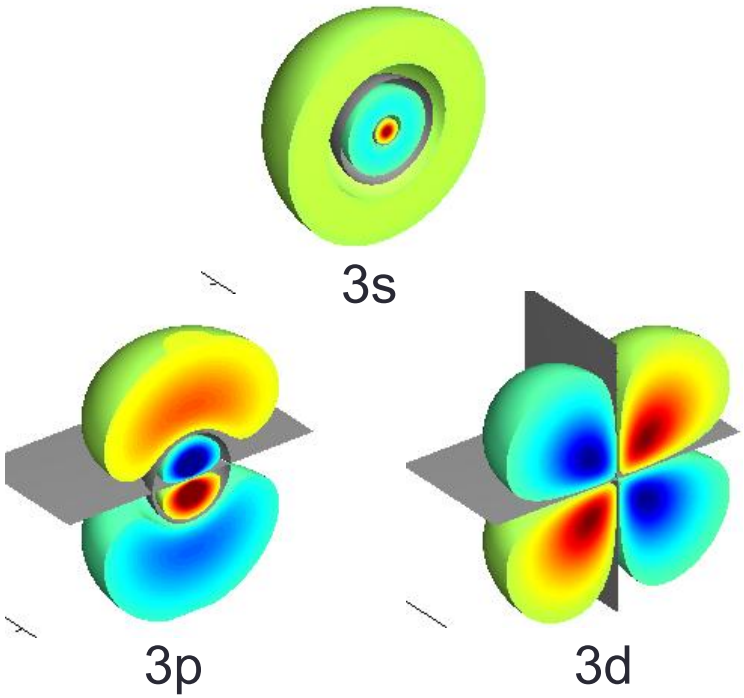
3*p*

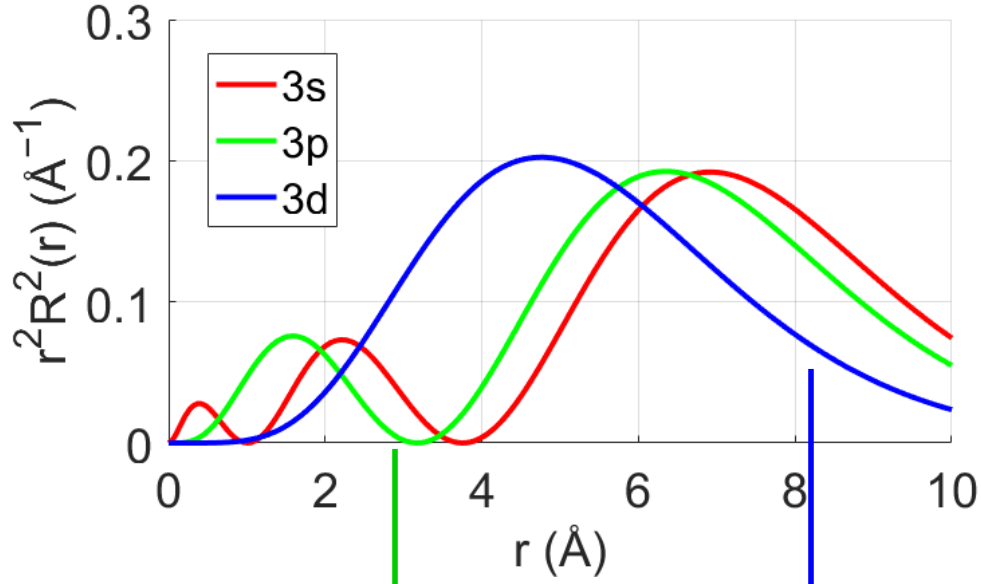


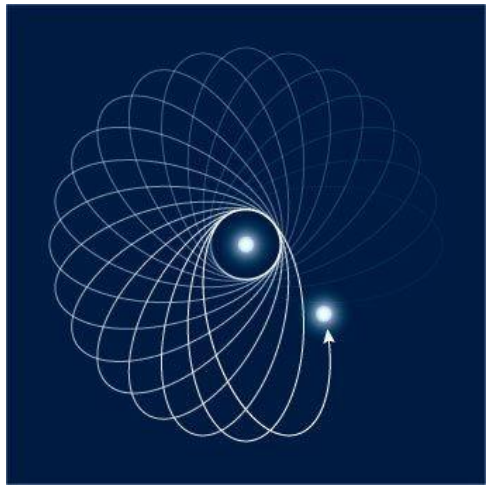
Elliptical orbits
椭圆轨道

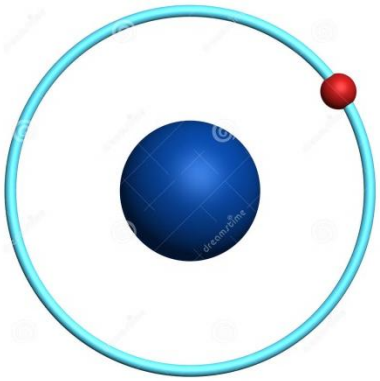
Radial Distribution

Penetration effect 钻穿效应









For H atom,

$r(3s) \neq r(3p) \neq r(3d)$

$E_k(3s) = E_k(3p) = E_k(3d)$

$E_p(3s) = E_p(3p) = E_p(3d)$

Summary

Principal quantum number n = Total number of nodes + 1

$n = 1, 2, 3, \ldots$

Angular momentum quantum number l = Number of nodal planes

$l = 0, 1, 2, \ldots, n - 1.$

$$E_n = -\frac{13.6\ \text{eV}}{n^2}$$

$$L = \underbrace{\sqrt{l(l+1)} \cdot h/2\pi}_{\text{3D}} \approx \underbrace{l \cdot h/2\pi}_{\text{2D}}$$

For 1*s*, 2*p*, 3*d*, …: $r_{\text{mp},n} = n^2 a_0$

2*s*, 3*s*, 2*p* exhibit penetration effect.